

CLASS TOPIC

Foundations, data and descriptive statistics

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

W01**Statistics**

Clean formulas · aligned layout · corrected te

CLASS TOPIC

Foundations, data and descriptive statistics

Economic question first

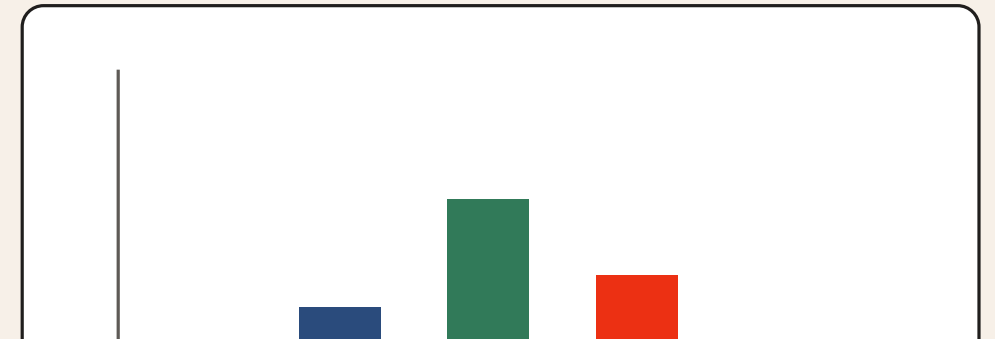
- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

FORMULA

$$\bar{x} = (1/n) \sum x_i$$

median = middle ordered value

$$\text{range} = \max(x) - \min(x)$$



CLASS TOPIC

Foundations, data and descriptive statistics

Key definitions

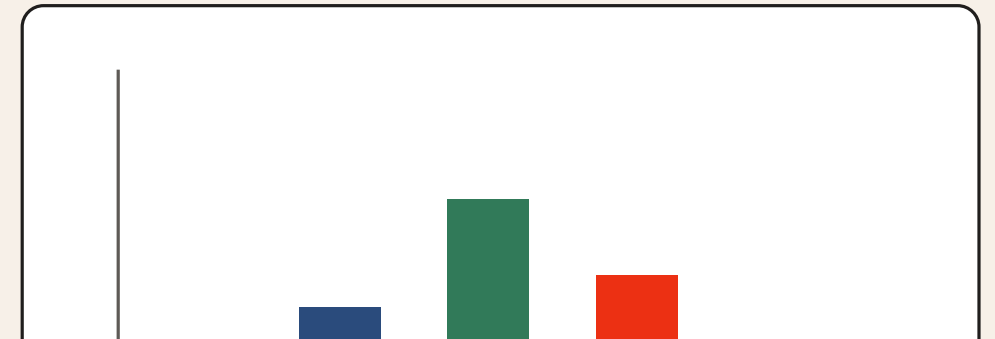
- Population: full group of interest.
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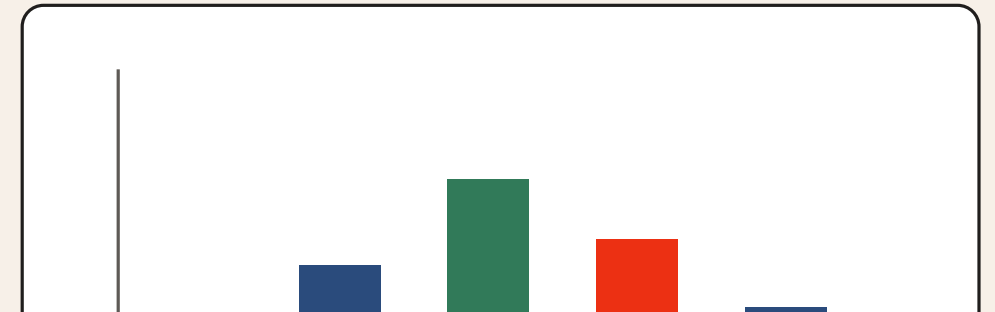
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Mini example

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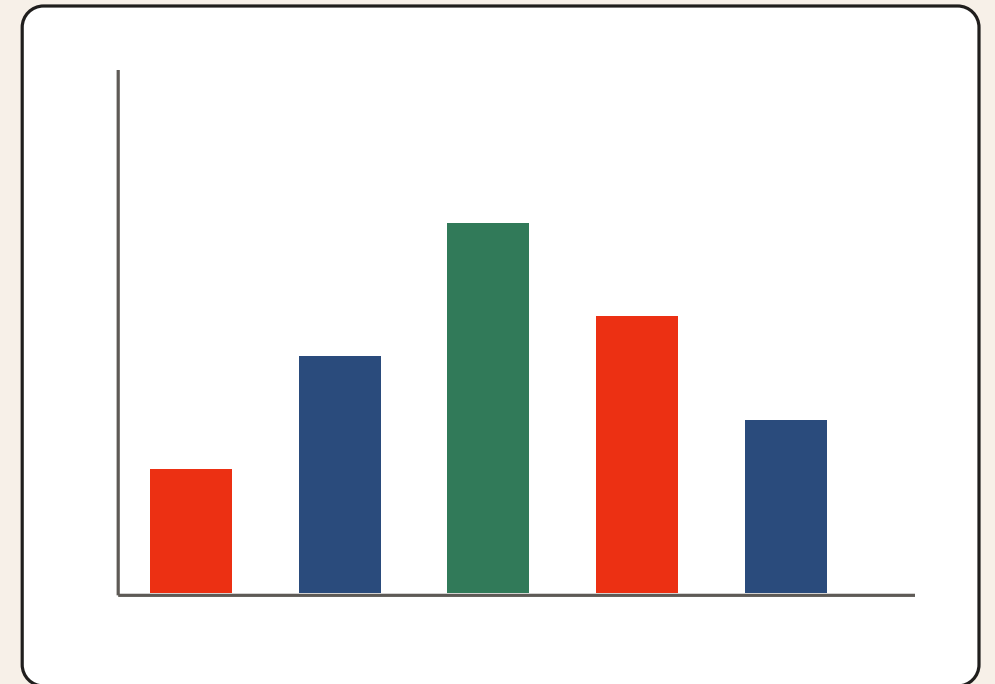
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Visual intuition

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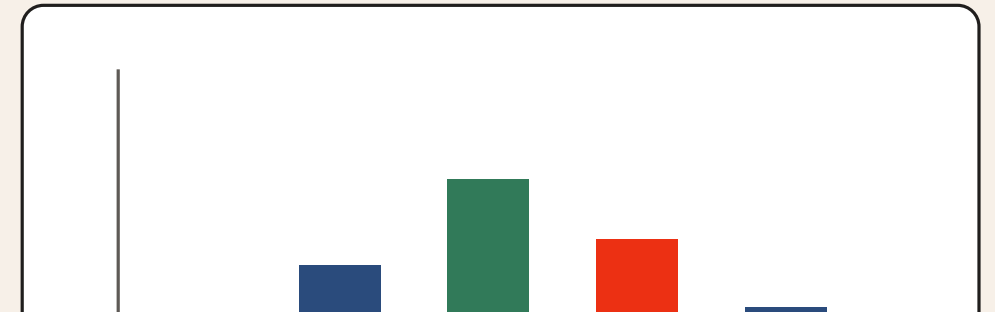
FORMULA 2

$$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

$$s = \sqrt{s^2}$$

$$CV = s / \bar{x}$$

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CLASS TOPIC

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Common mistakes

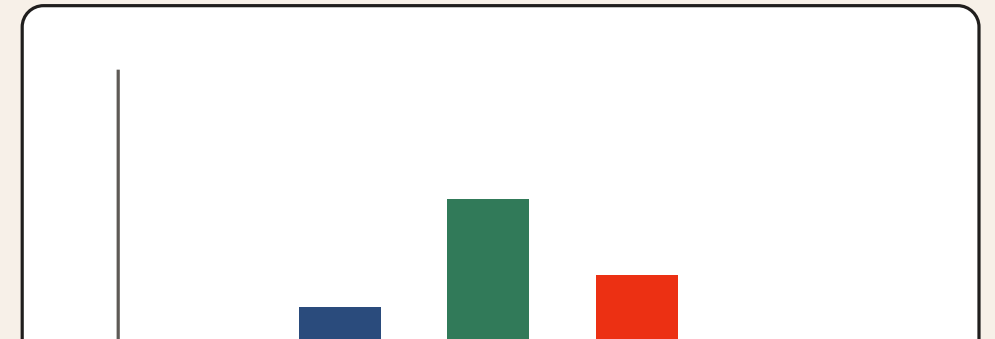
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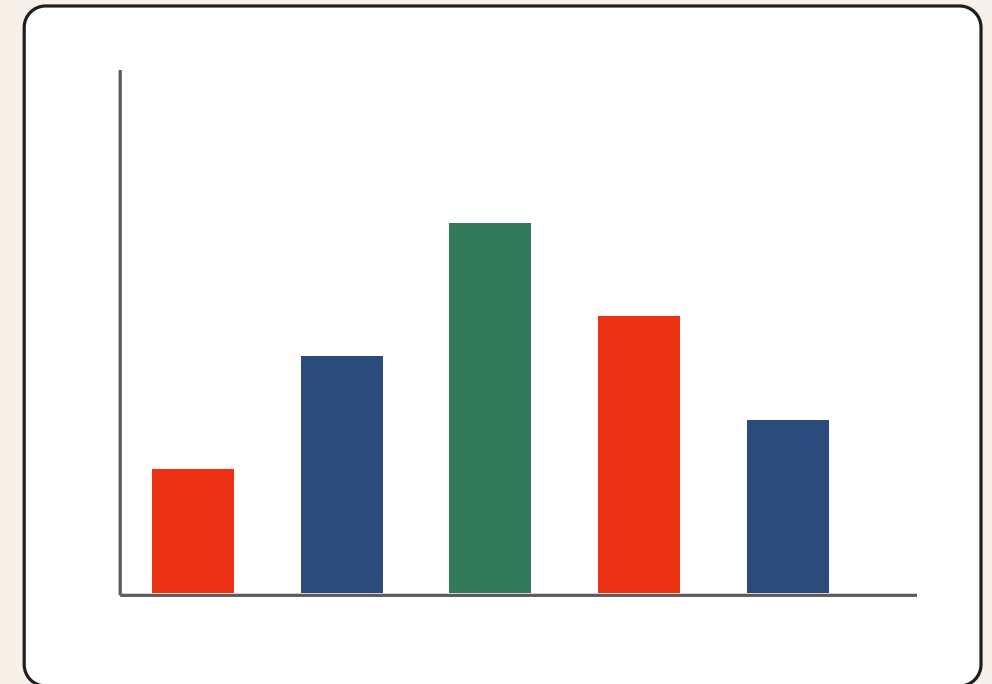
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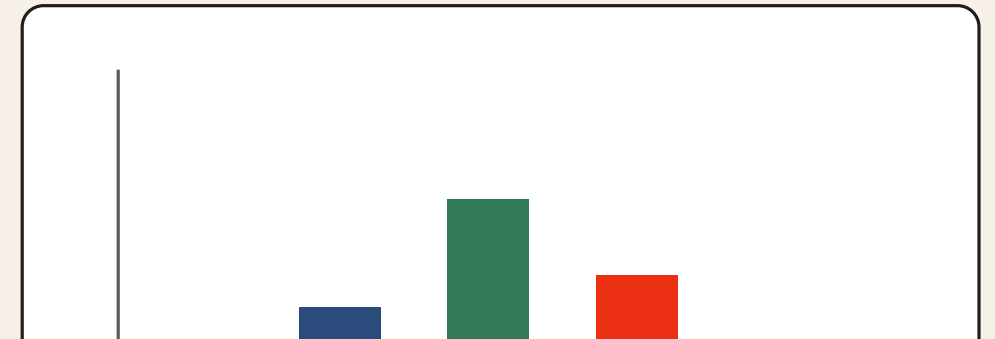
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Class exercise

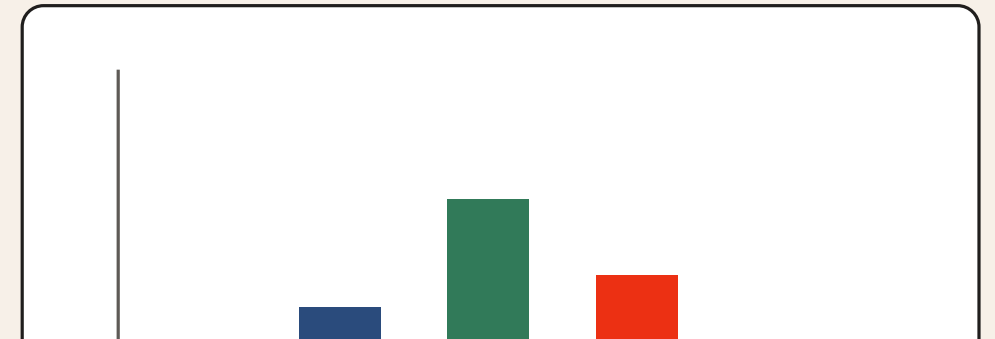
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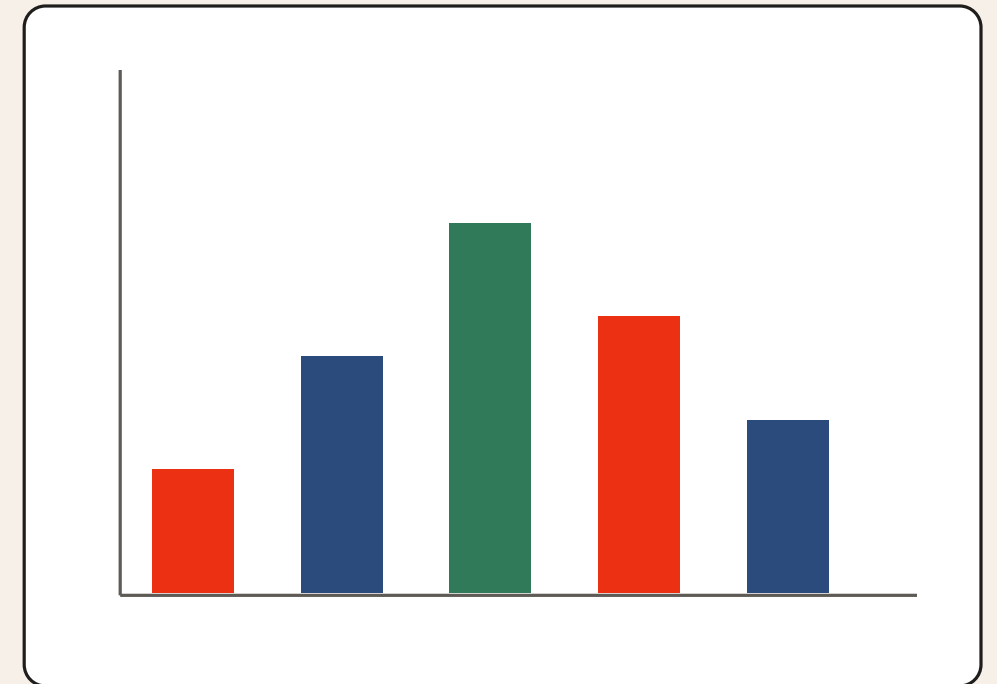


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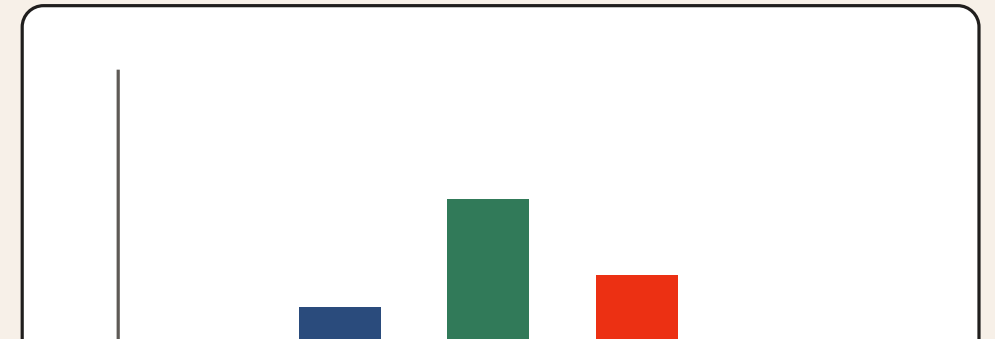
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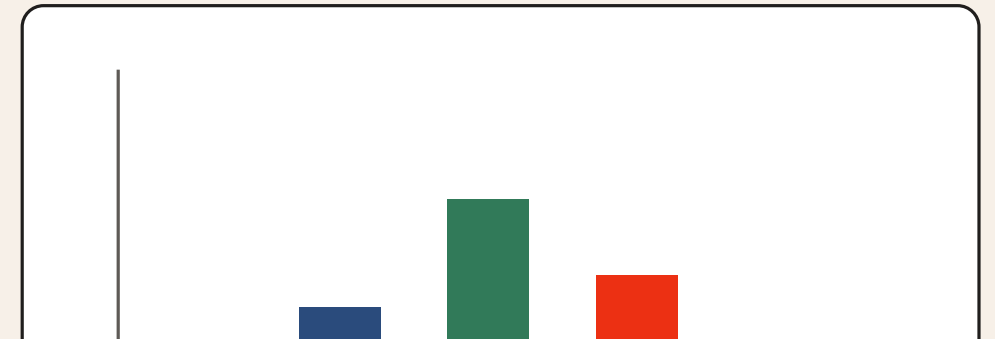
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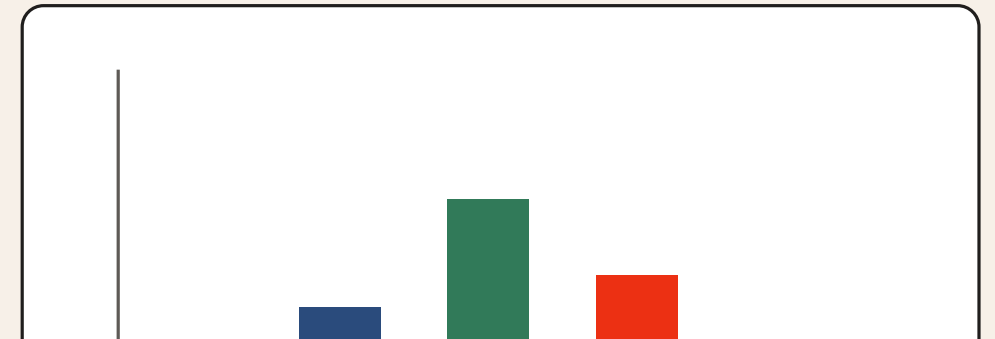
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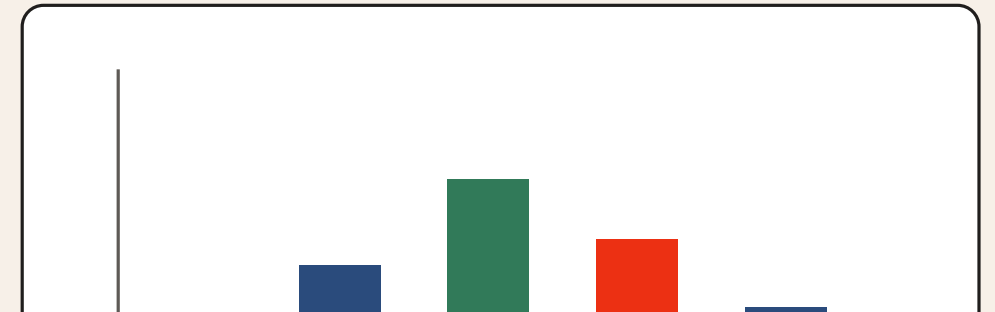
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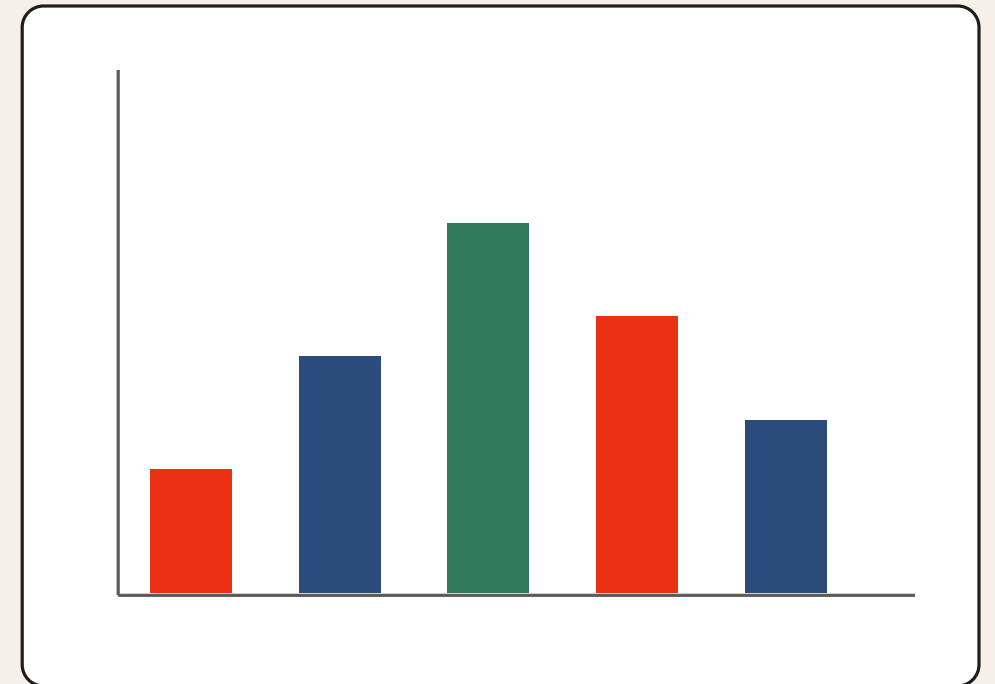
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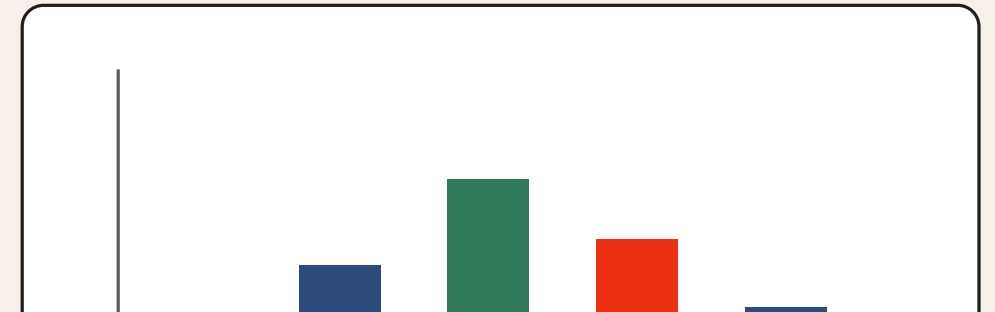
FORMULA 2

$$z_i = (x_i - \bar{x}) / s$$

$$\text{IQR} = Q_3 - Q_1$$

$$\text{outlier if } x < Q_1 - 1.5\text{IQR} \text{ or } x > Q_3 + 1.5\text{IQR}$$

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Common mistakes

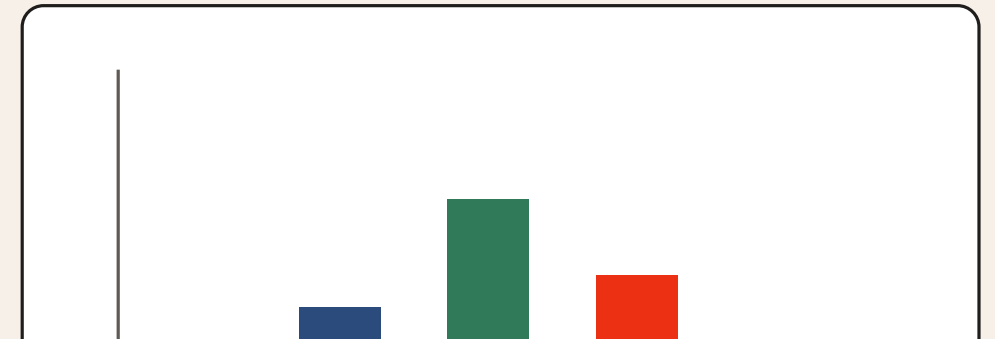
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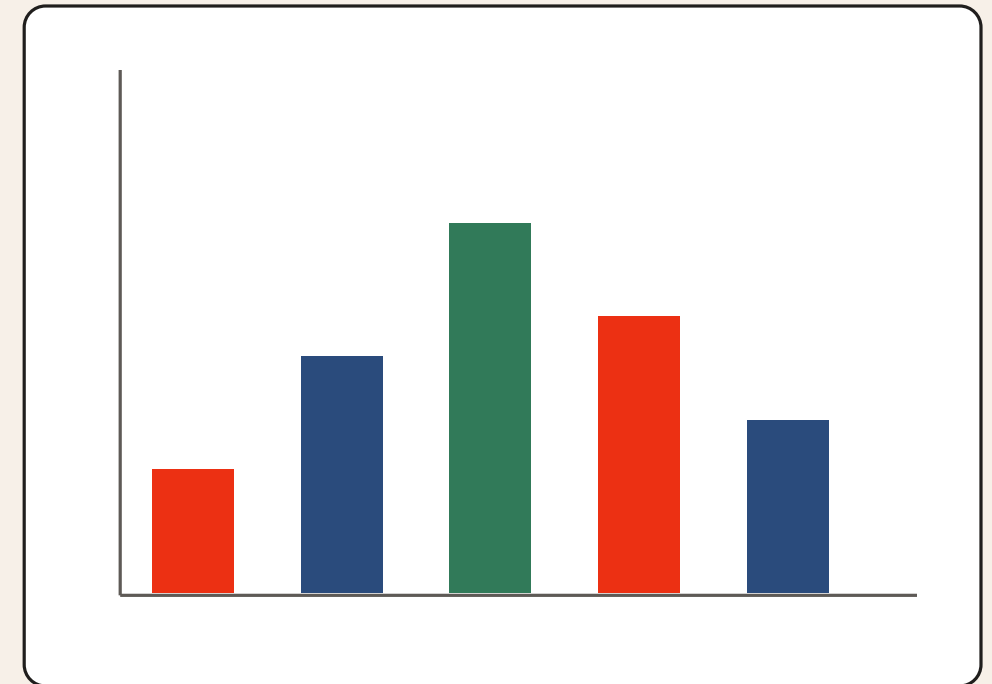
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Conditions and assumptions

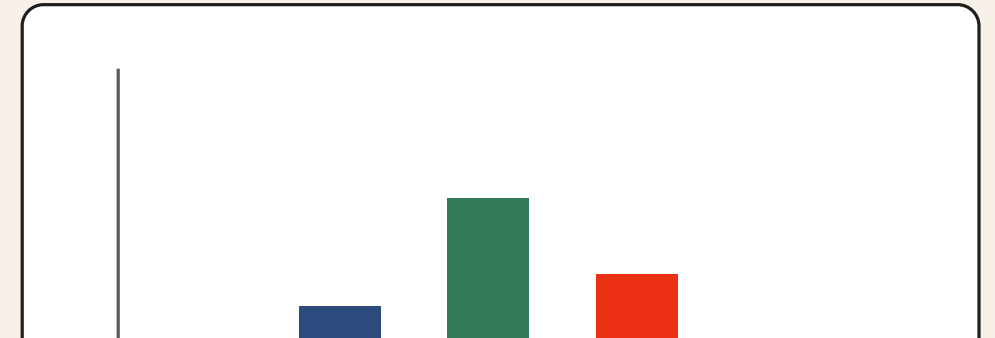
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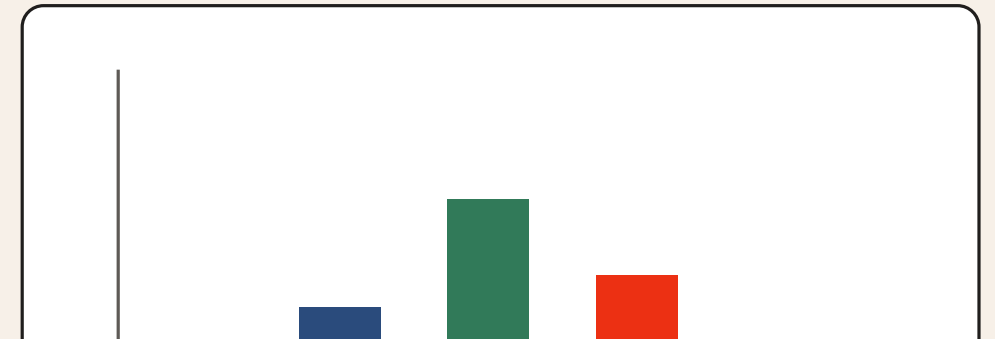
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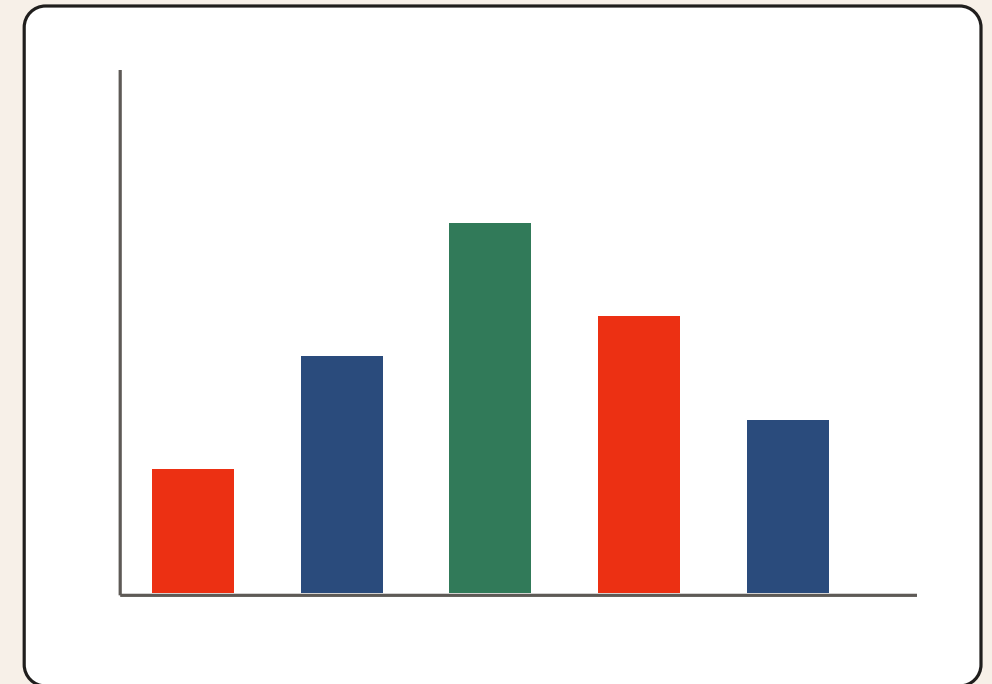


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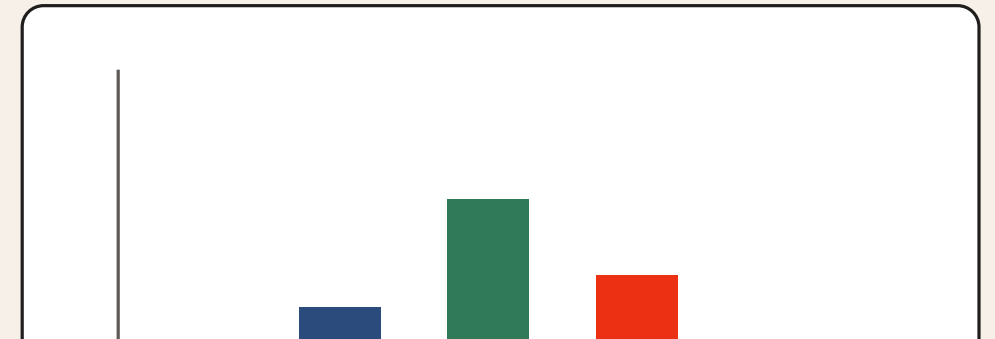
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W03**Statistics**

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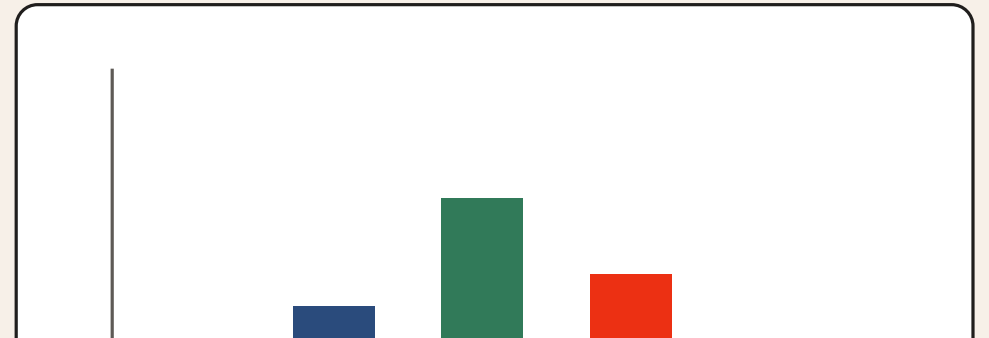
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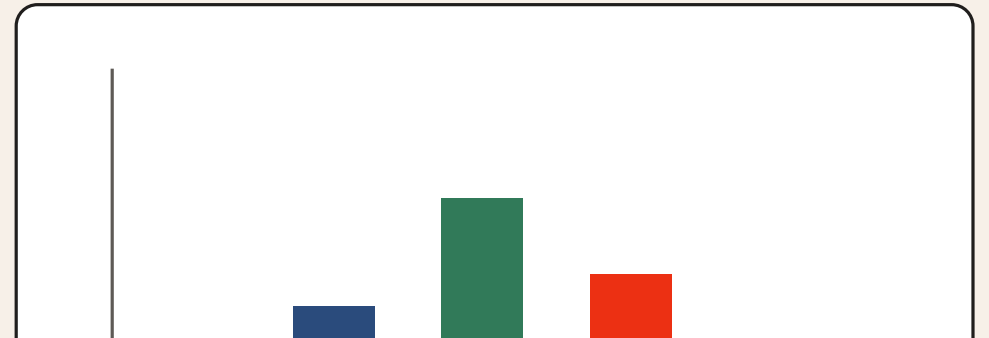
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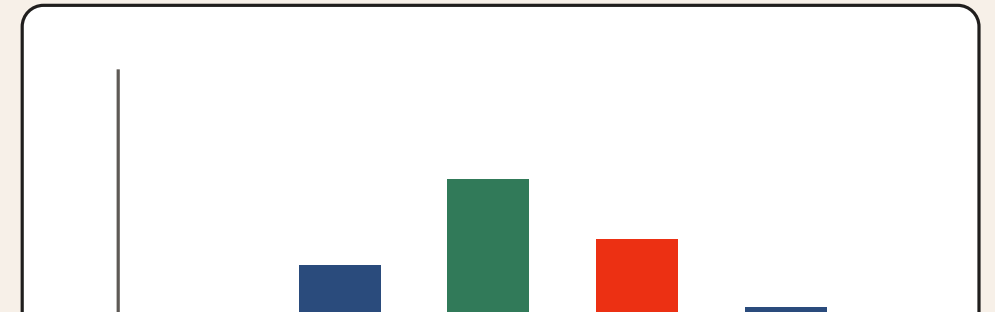
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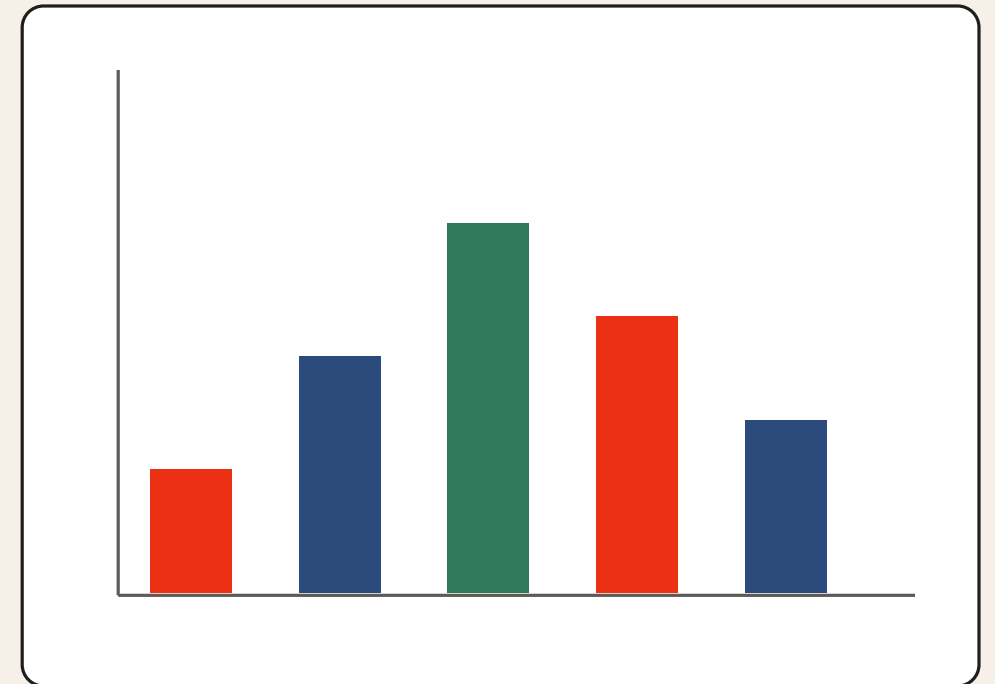
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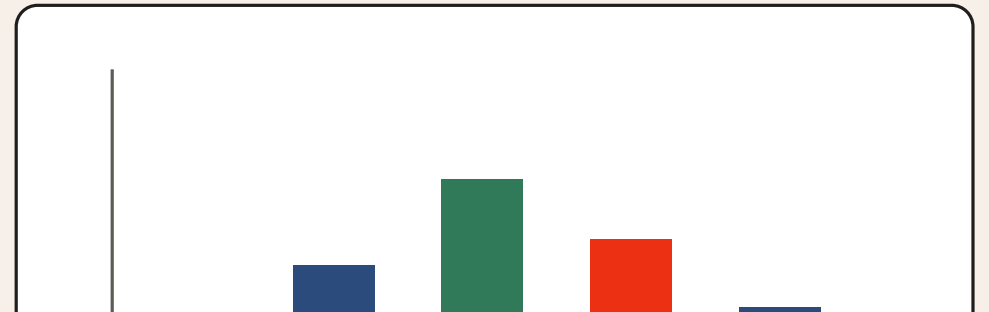
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FORMULA 2

$$P(A | B) = P(A \cap B) / P(B)$$

$$P(A | B) = P(B | A)P(A) / P(B)$$



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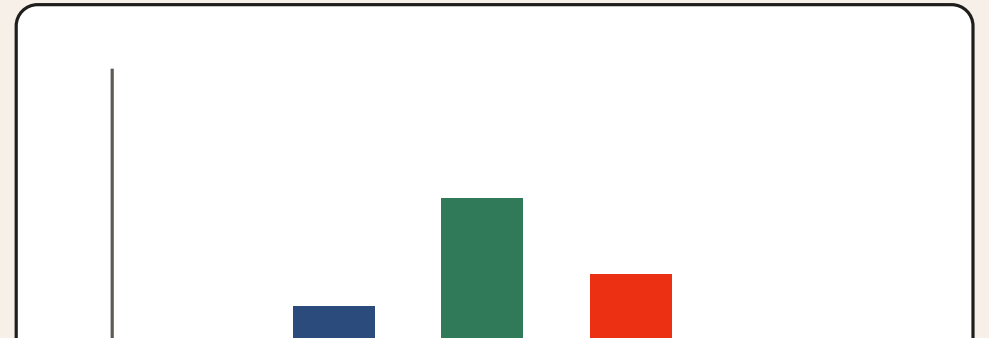
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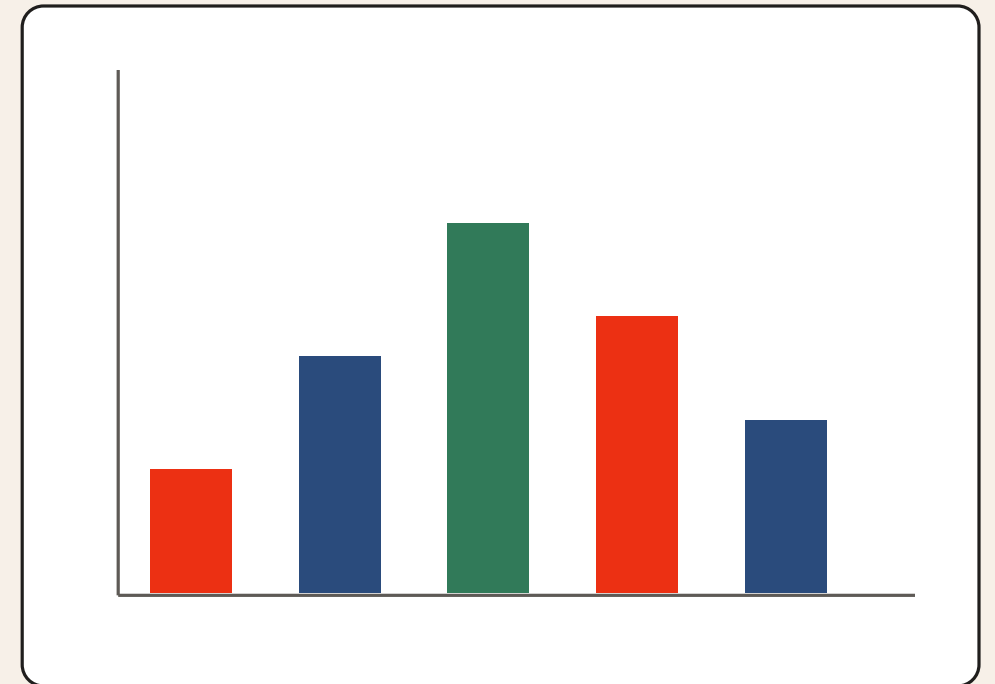
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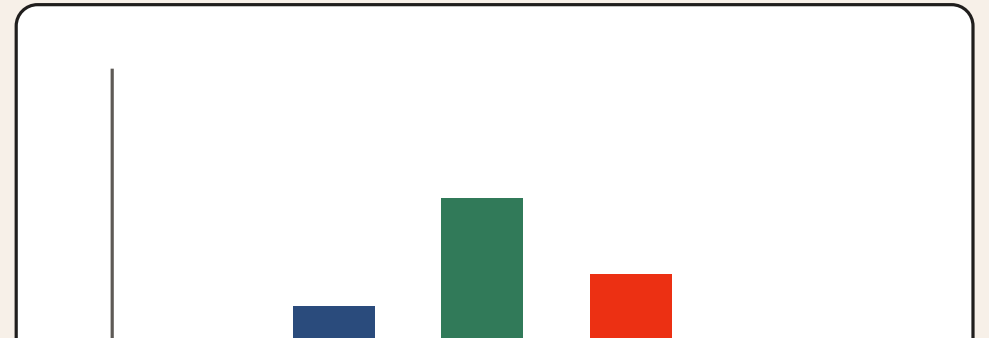
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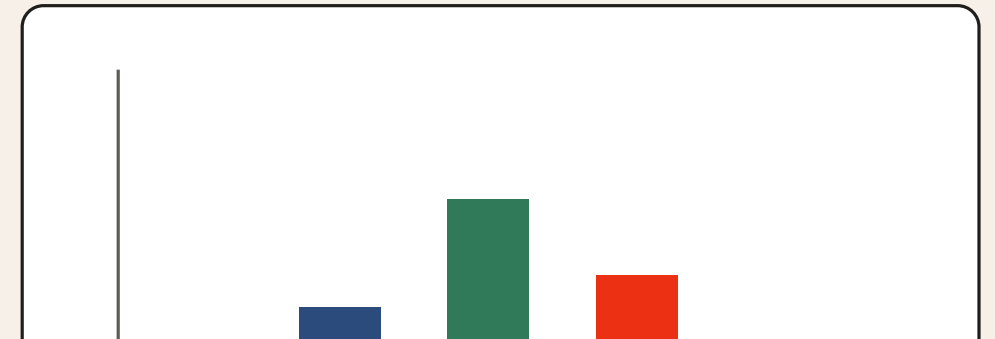
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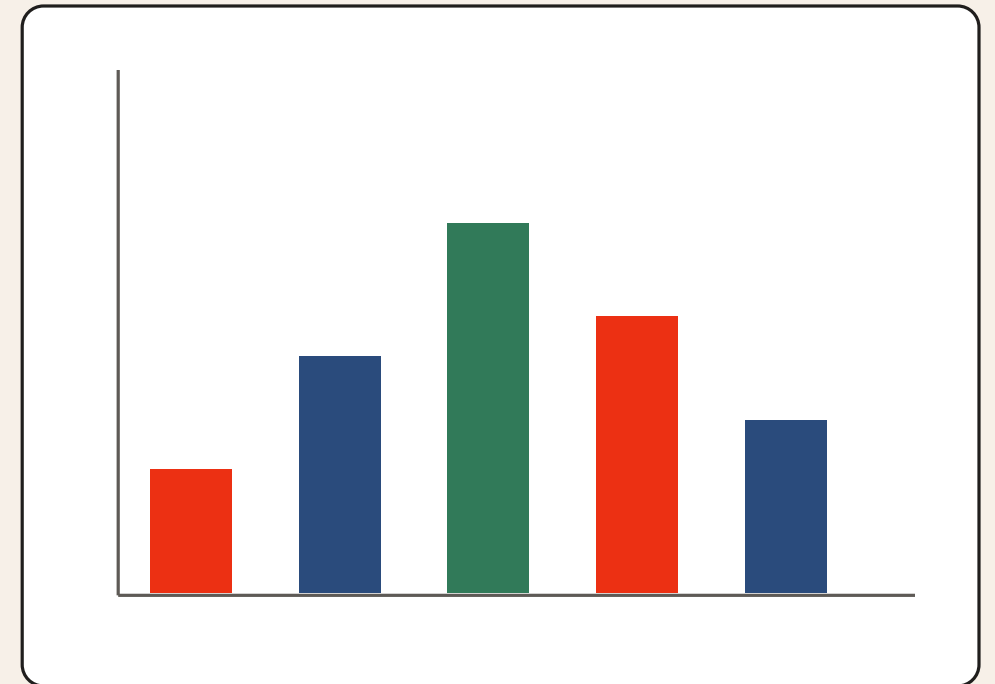


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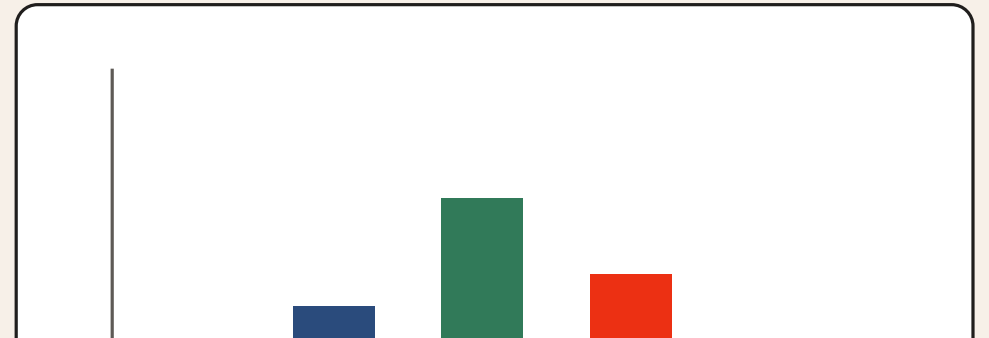
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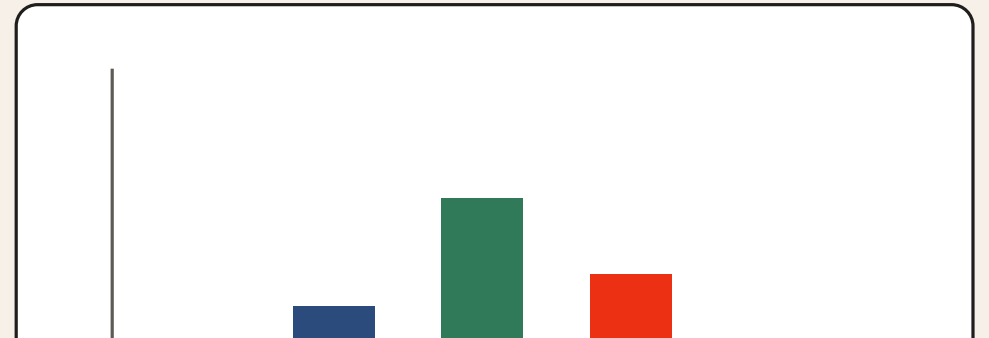
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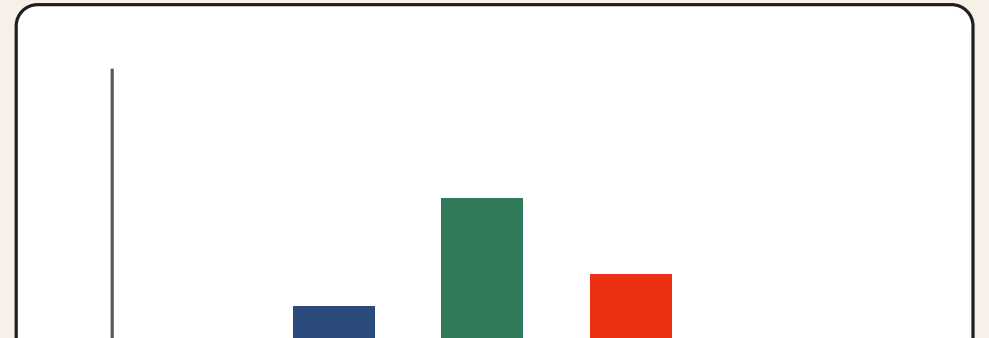
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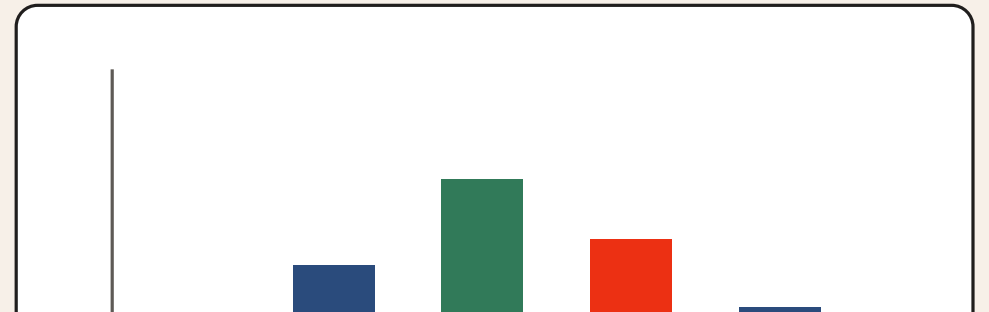
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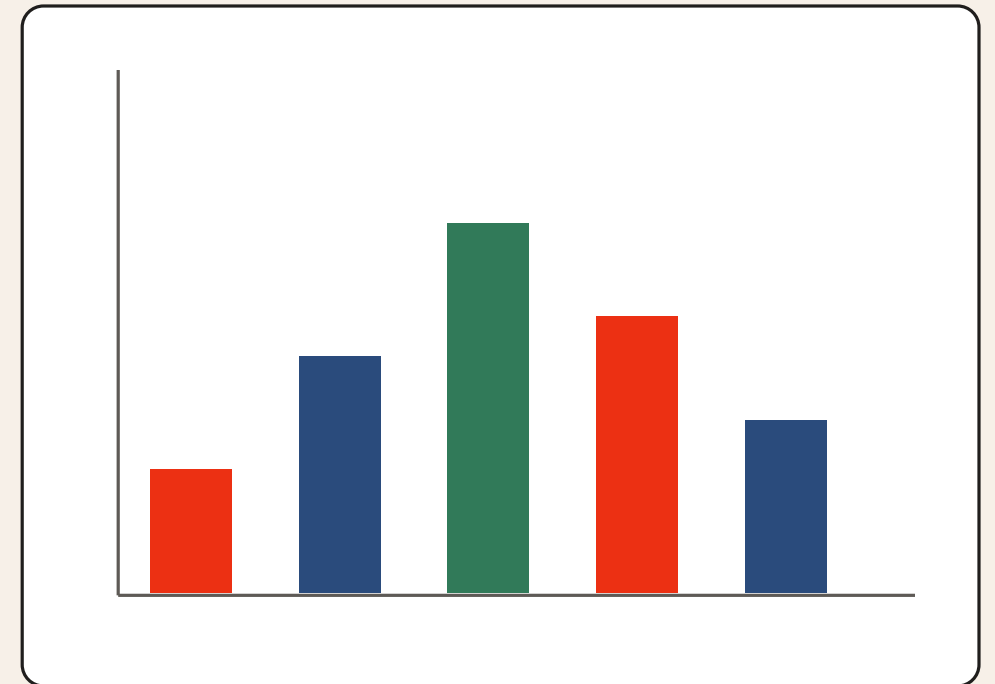
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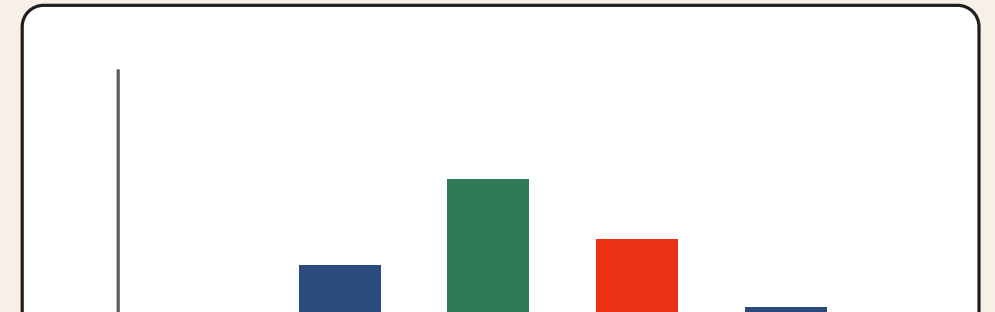
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FORMULA 2

Binomial: $P(X = k) = C(n, k)p^k(1 - p)^{n-k}$
 $E(X) = np, \text{ Var}(X) = np(1 - p)$



CLASS TOPIC

Random variables and discrete distributions

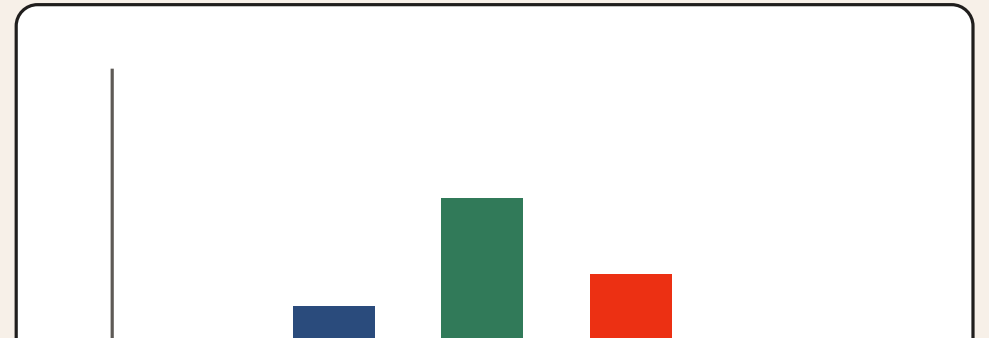
Common mistakes

- Mixing sample and population symbols.
- Using a formula without checking conditions.
- Reporting a number without units or interpretation.

FORMULA

$$E(X) = \sum x \cdot P(X = x)$$

$$\text{Var}(X) = E[(X - \mu)^2] = \sum (x - \mu)^2 p(x)$$



CLASS TOPIC

Random variables and discrete distributions

R / spreadsheet check

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R / EXCEL

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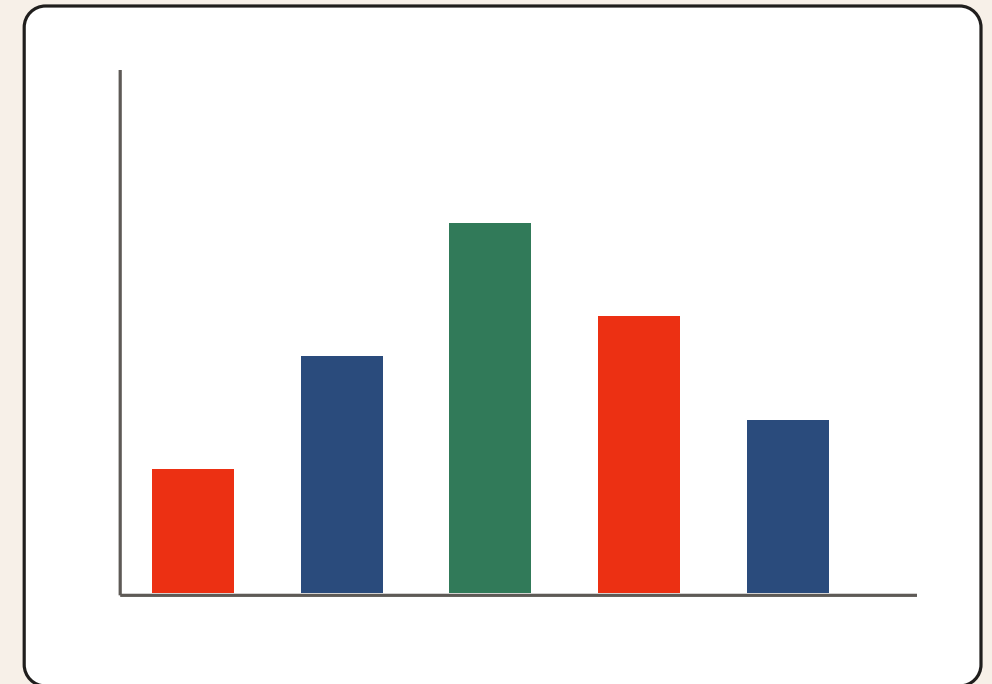
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CLASS TOPIC

Random variables and discrete distributions

Interpretation sentence

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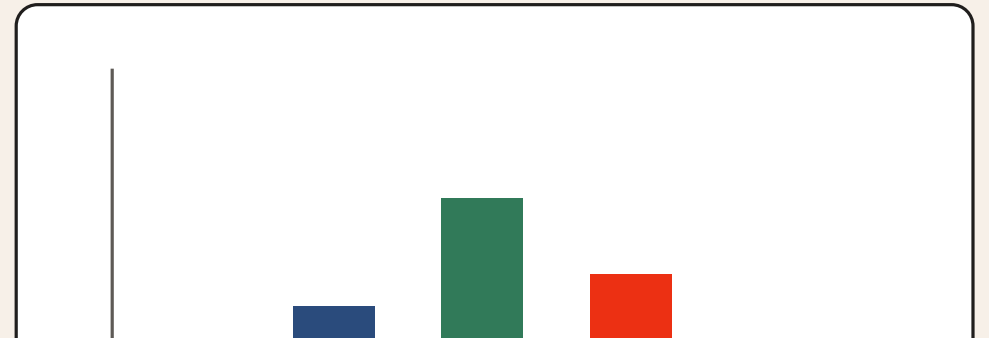
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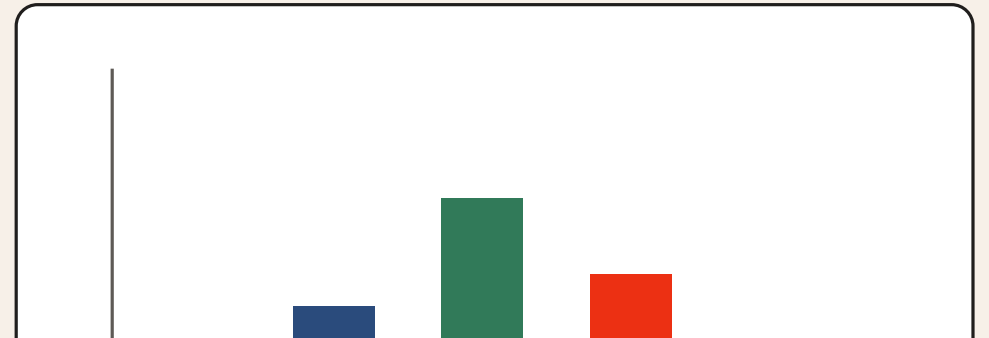
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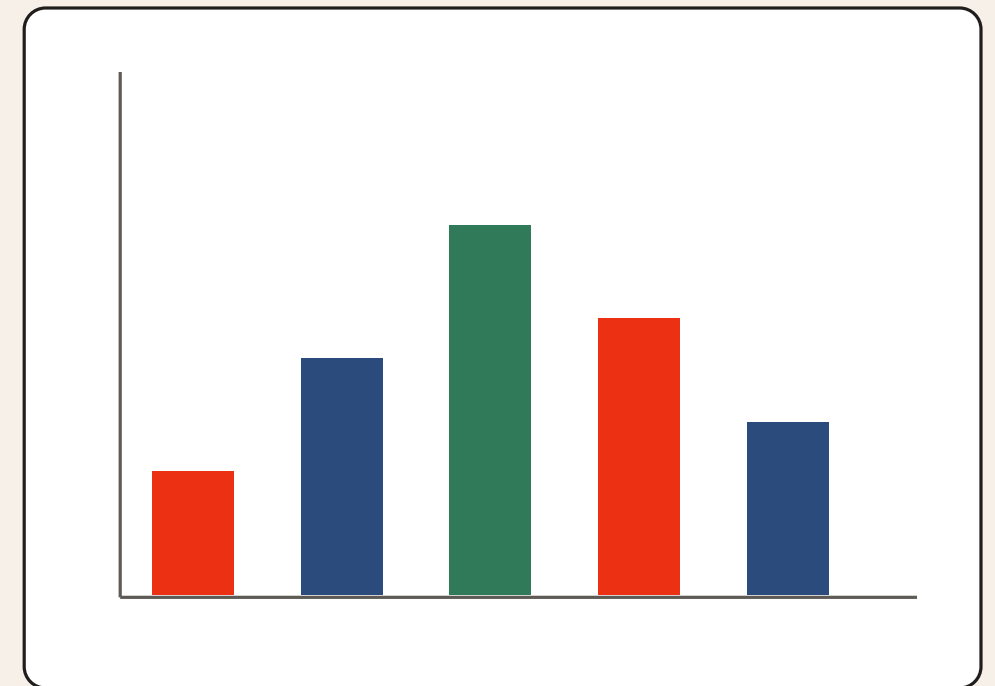


CLASS TOPIC

Random variables and discrete distributions

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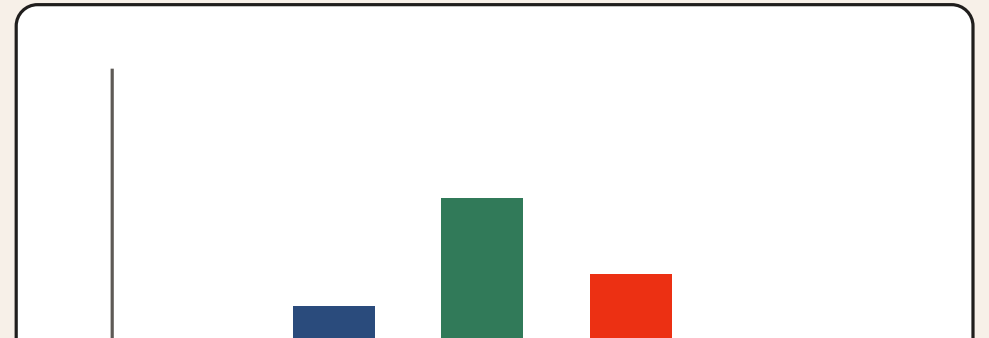
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CLASS TOPIC

Continuous and normal distributions

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
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W05**Statistics**

Clean formulas · aligned layout · corrected te

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



CLASS TOPIC

Continuous and normal distributions

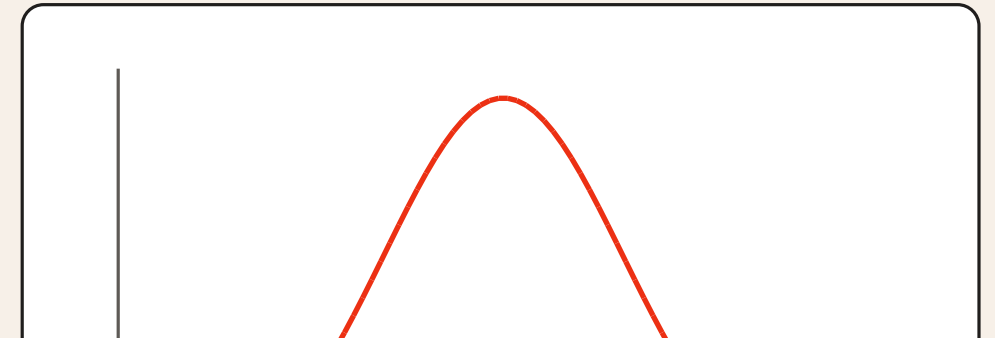
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Total area under $f(x)$ equals 1



CLASS TOPIC

Continuous and normal distributions

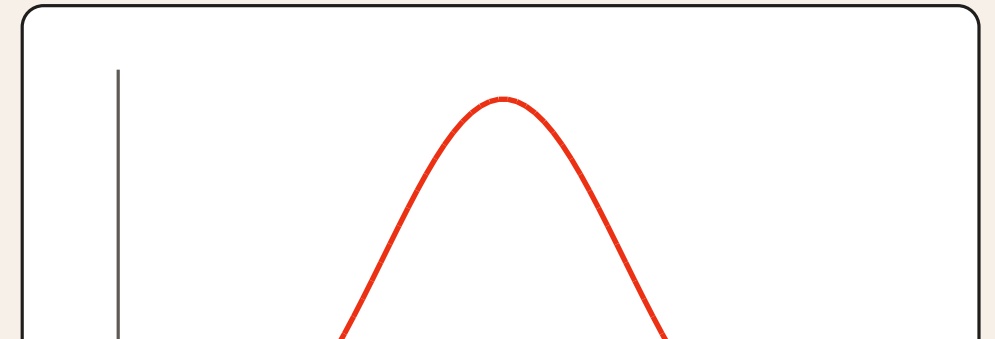
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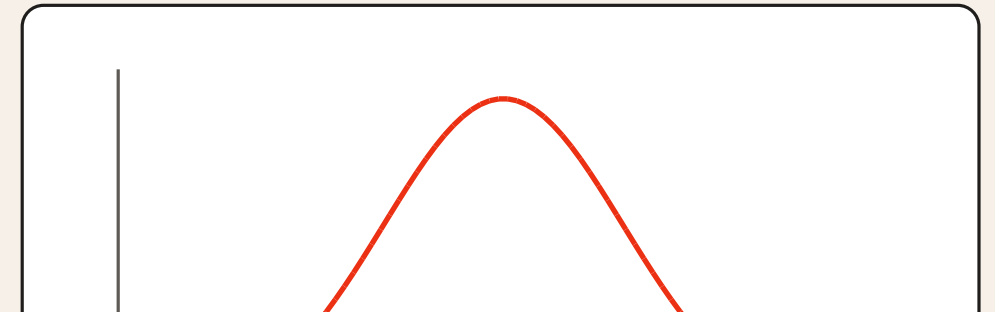
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CLASS TOPIC

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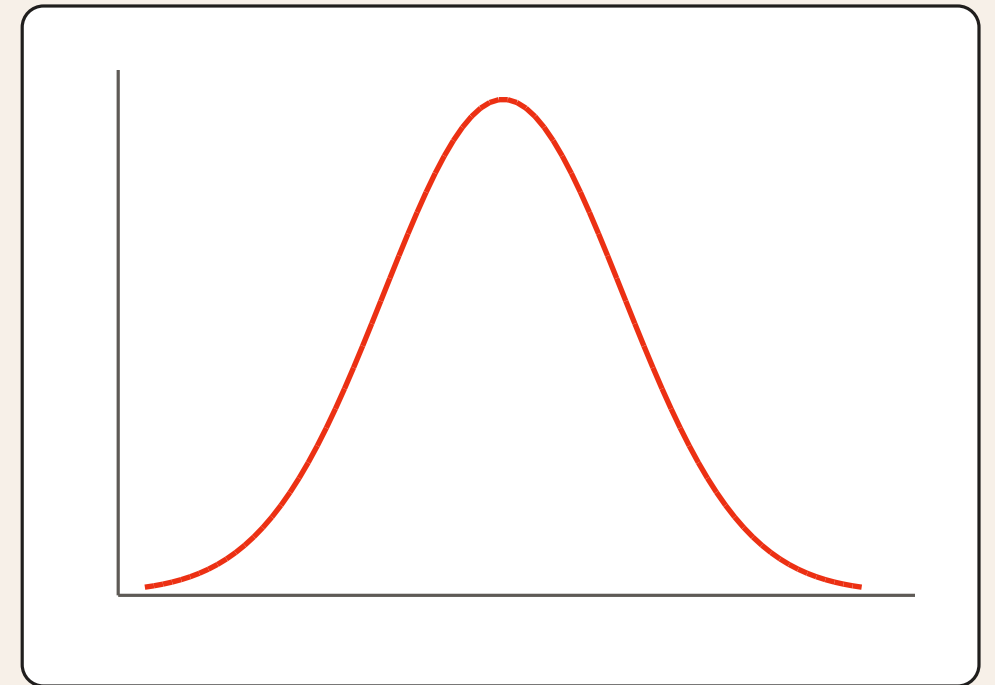
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CLASS TOPIC

Continuous and normal distributions

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CLASS TOPIC

Continuous and normal distributions

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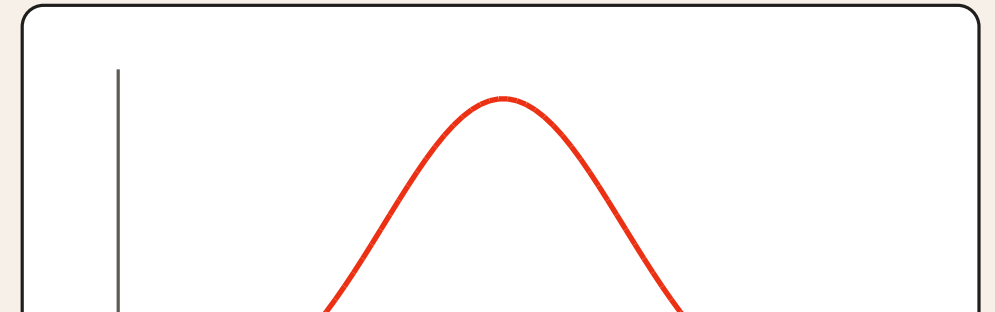
FORMULA 2

$$Z = (X - \mu) / \sigma$$

$$X \sim N(\mu, \sigma^2)$$

$$P(X \leq x) = P(Z \leq (x - \mu) / \sigma)$$

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



CLASS TOPIC

Continuous and normal distributions

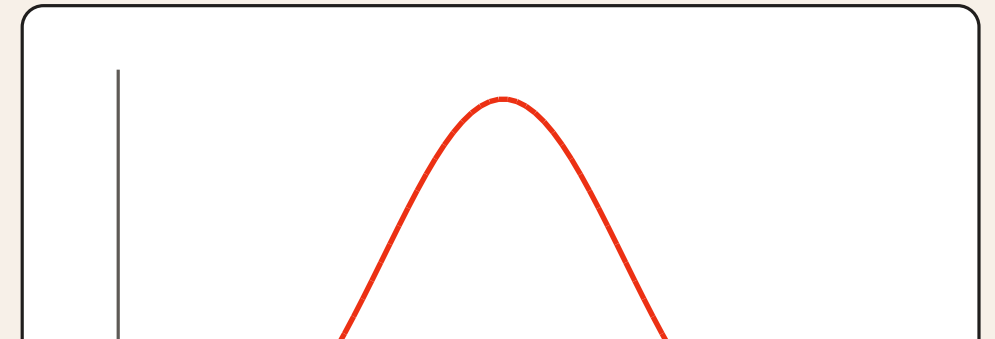
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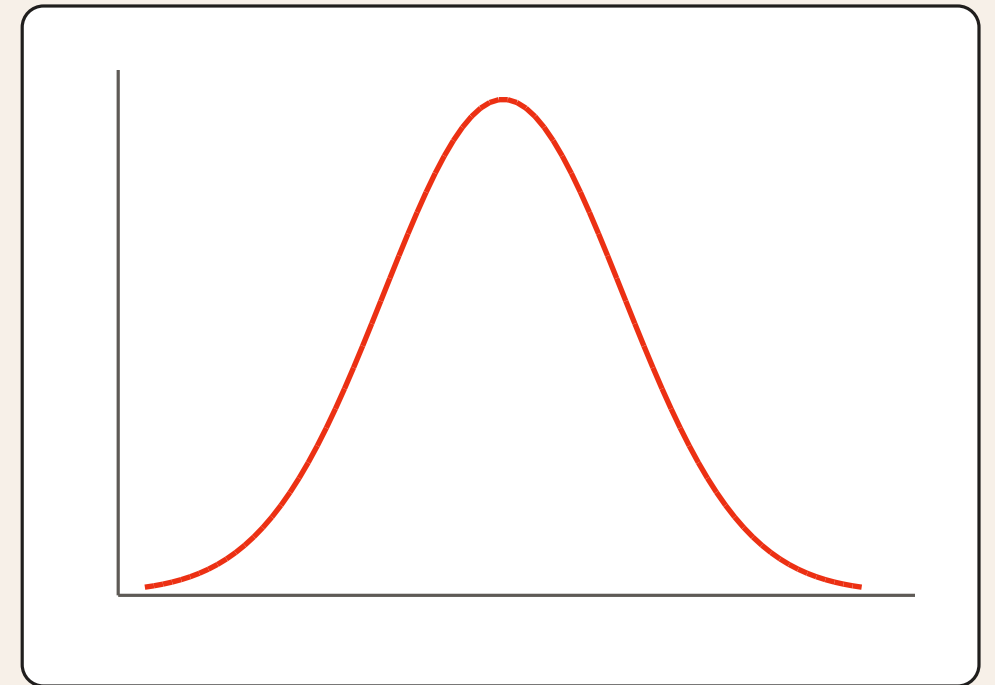
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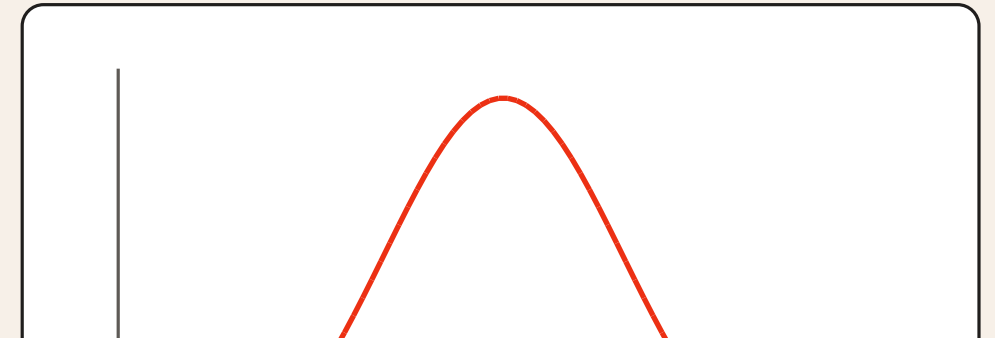
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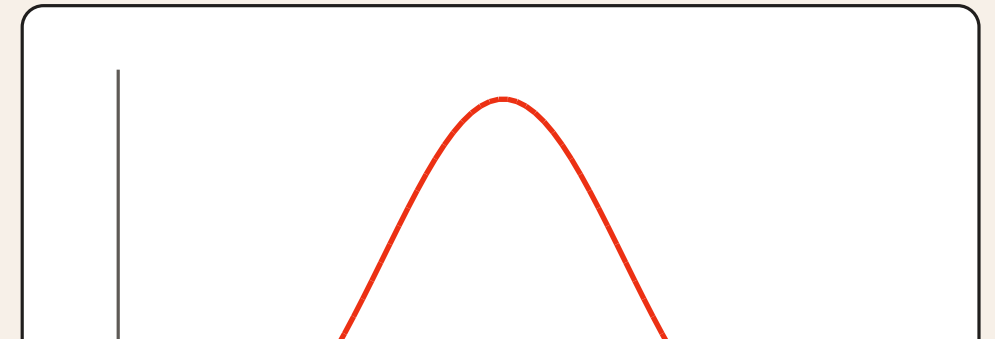
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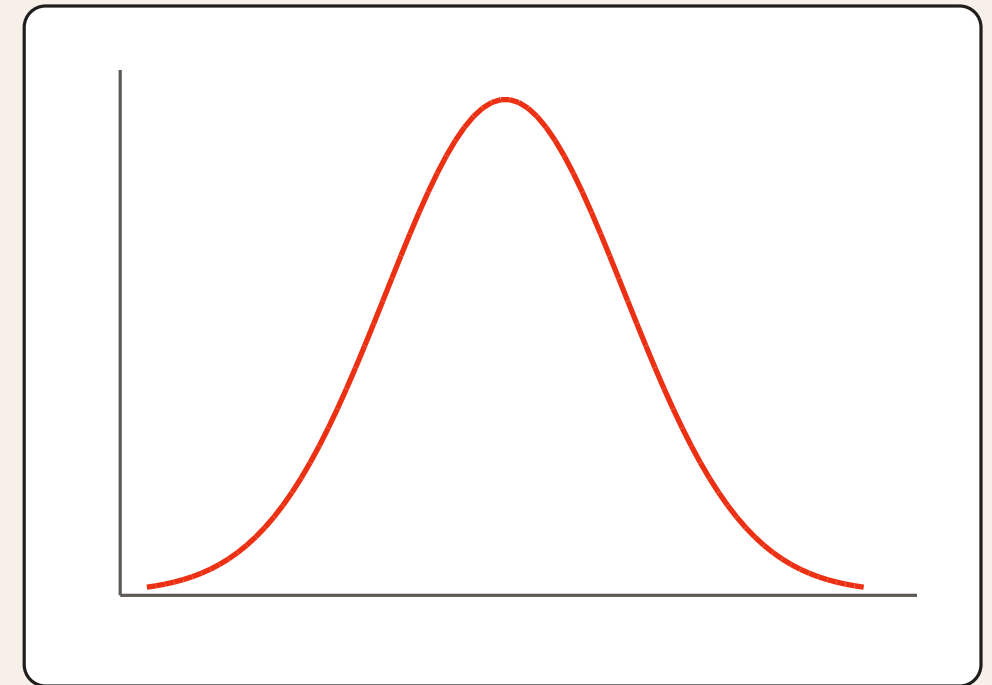


CLASS TOPIC

Continuous and normal distributions

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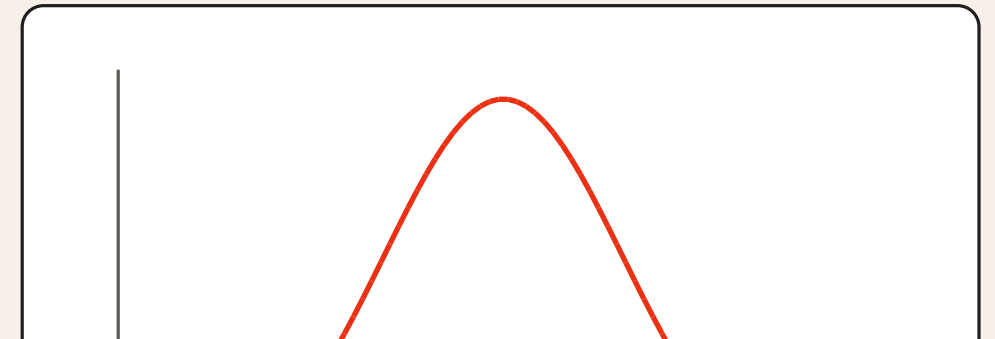
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CLASS TOPIC

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W06**Statistics**

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CLASS TOPIC

Joint distributions, covariance and correlation

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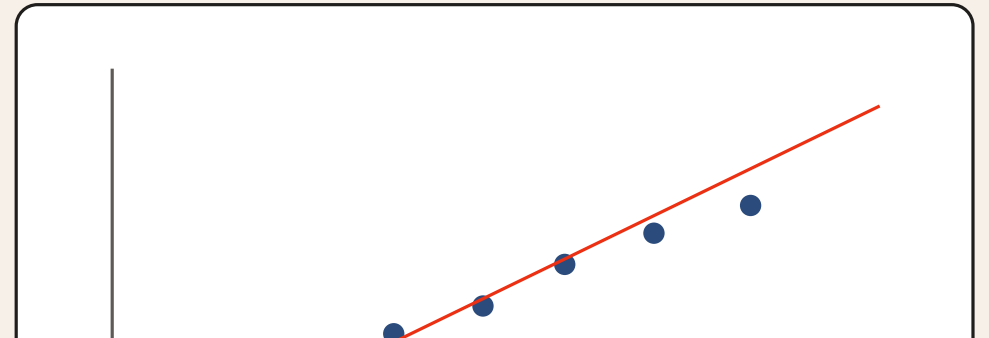
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CLASS TOPIC

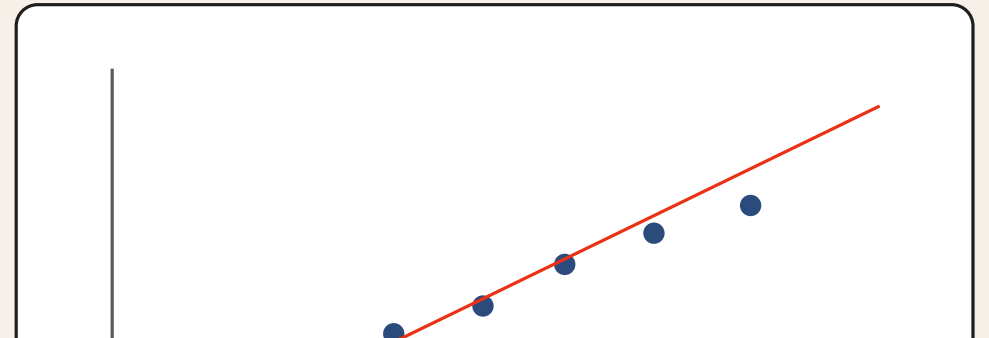
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CLASS TOPIC

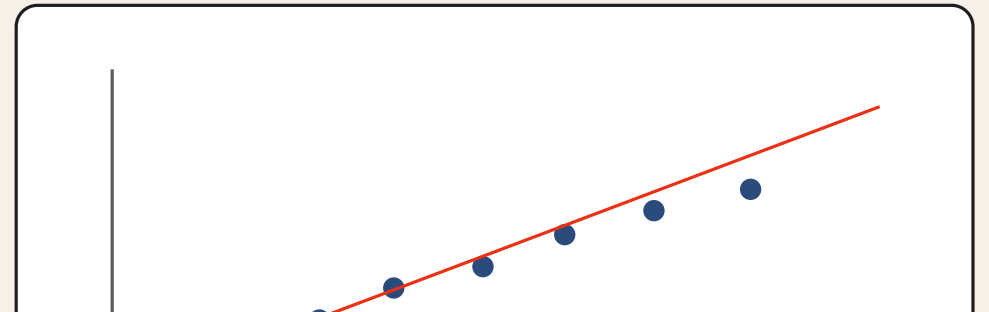
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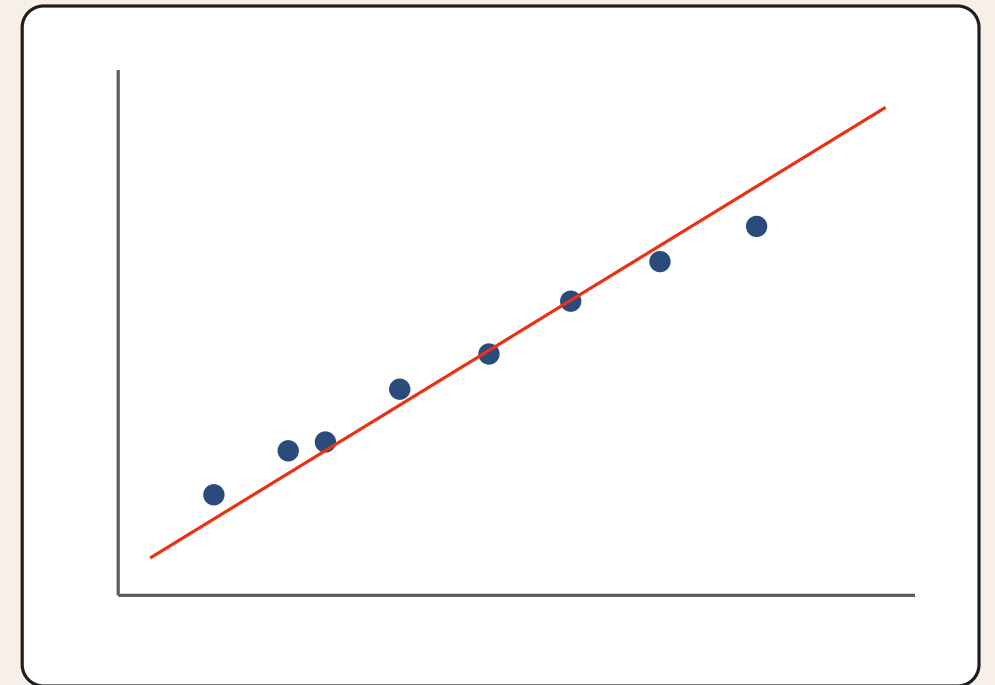
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CLASS TOPIC

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CLASS TOPIC

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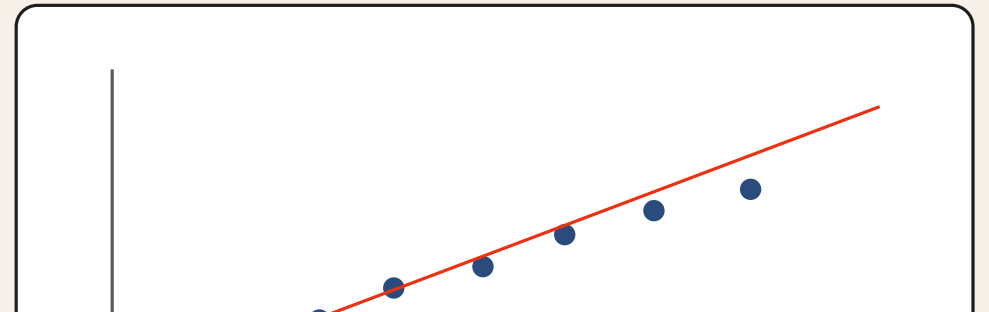
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FORMULA 2

$$\rho = \text{Cov}(X, Y) / (\sigma_x \sigma_y)$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2]}}$$

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CLASS TOPIC

Joint distributions, covariance and correlation

Common mistakes

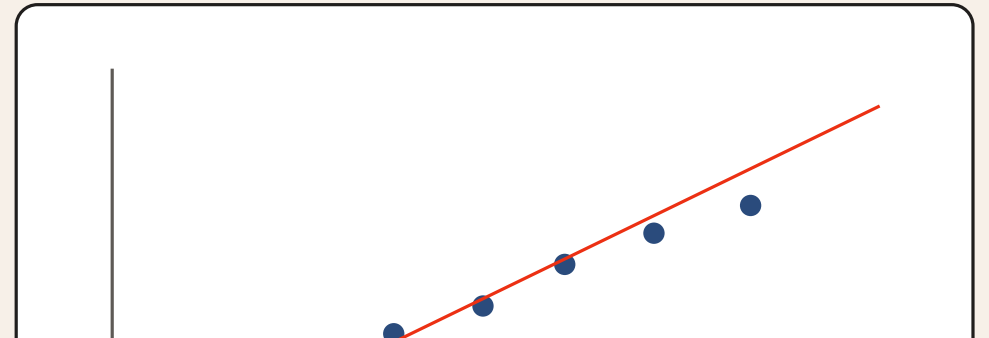
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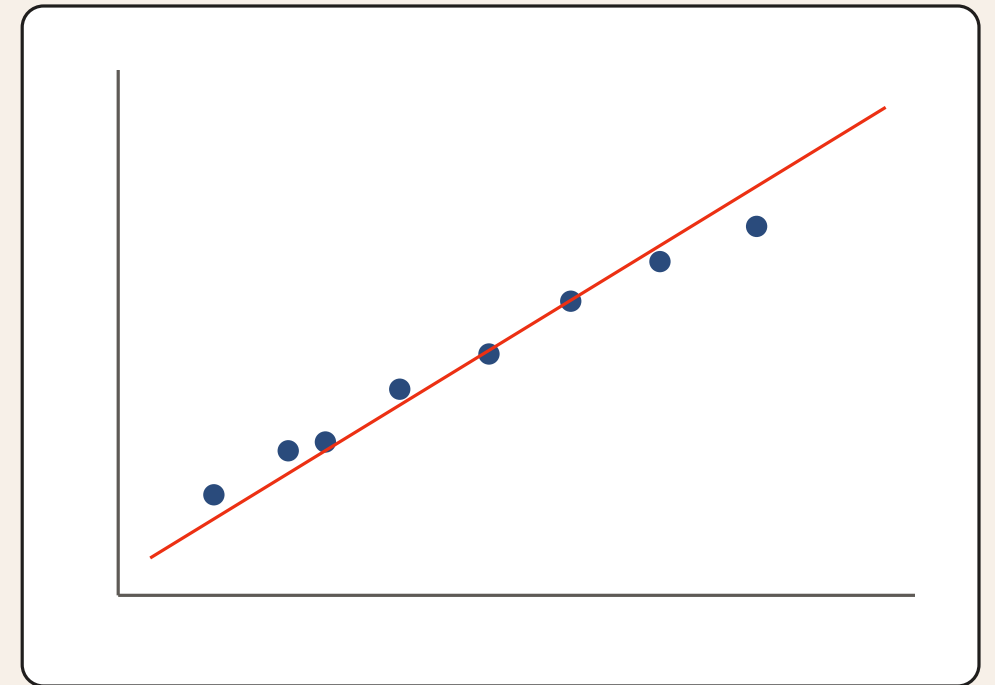
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CLASS TOPIC

Joint distributions, covariance and correlation

Interpretation sentence

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CLASS TOPIC

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Conditions and assumptions

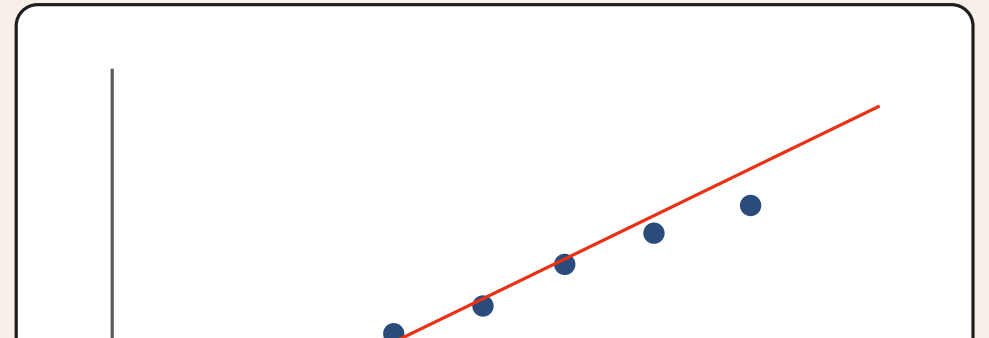
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CLASS TOPIC

Joint distributions, covariance and correlation

Class exercise

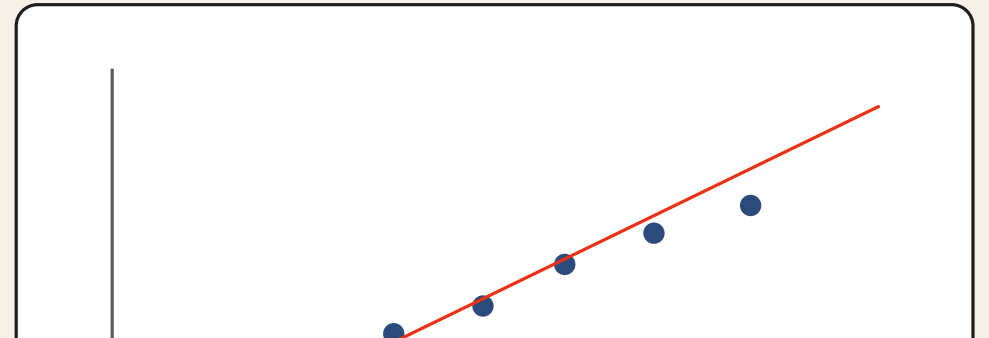
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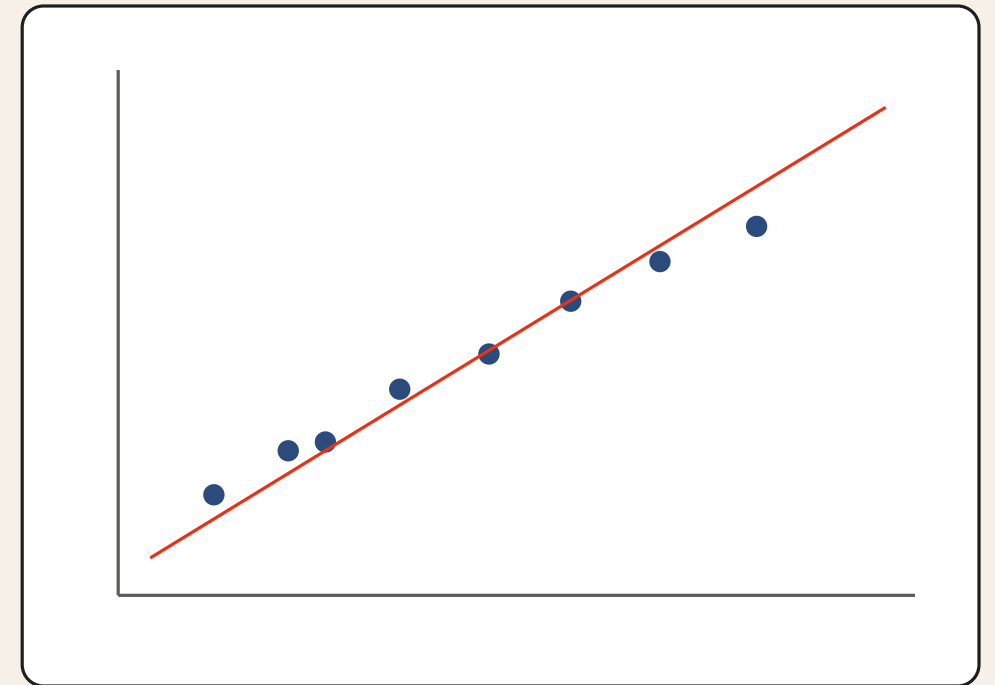


CLASS TOPIC

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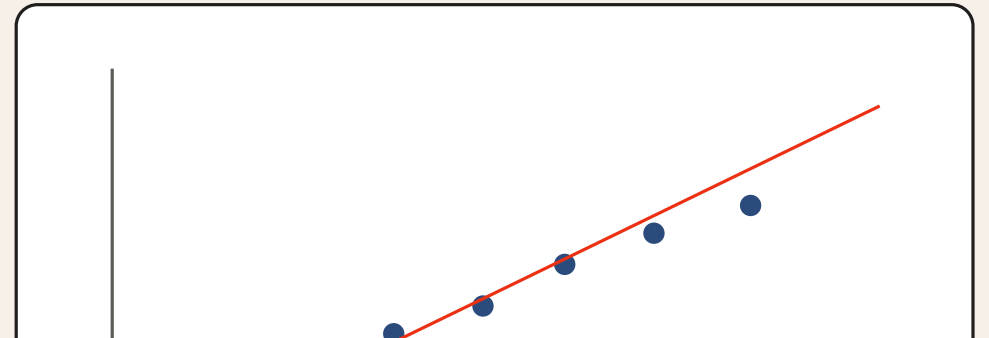
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CLASS TOPIC

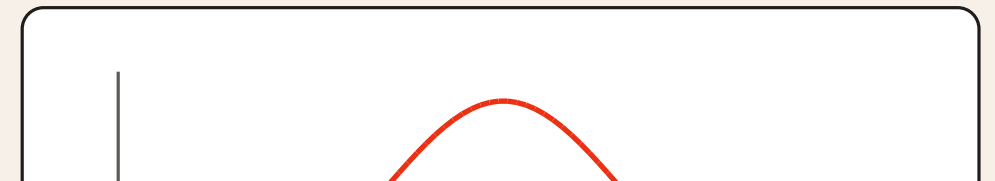
Sampling distributions and the central limit theorem

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W07**Statistics**

Clean formulas · aligned layout · corrected te



CLASS TOPIC

Sampling distributions and the central limit theorem

Economic question first

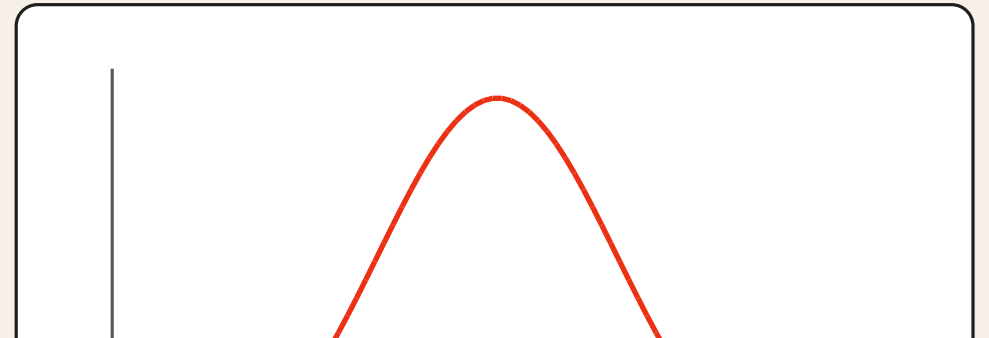
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FORMULA

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CLASS TOPIC

Sampling distributions and the central limit theorem

Key definitions

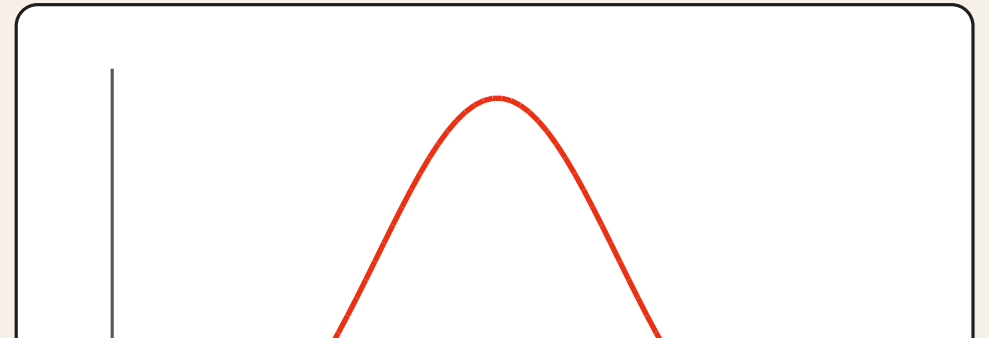
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CLASS TOPIC

Sampling distributions and the central limit theorem

Main formula box

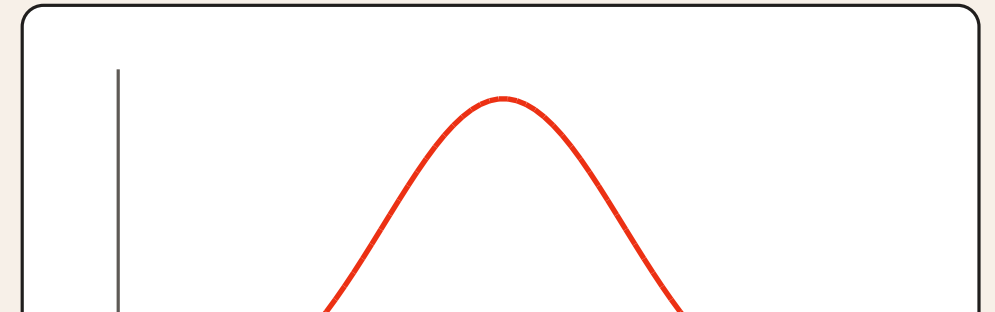
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CLASS TOPIC

Sampling distributions and the central limit theorem

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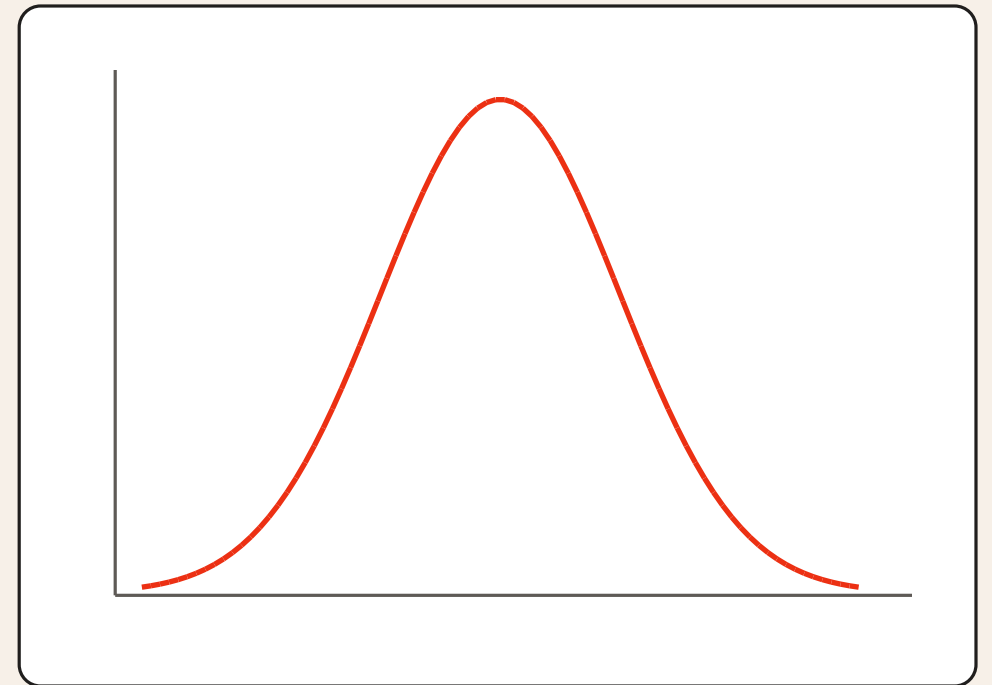
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CLASS TOPIC

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CLASS TOPIC

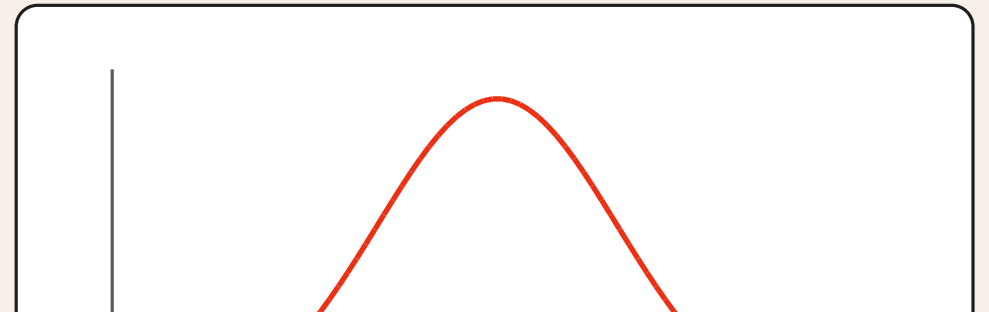
Sampling distributions and the central limit theorem

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FORMULA 2

$$\text{CLT: } \bar{X} \approx N(\mu, \sigma^2/n) \text{ for large } n$$
$$z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$$



CLASS TOPIC

Sampling distributions and the central limit theorem

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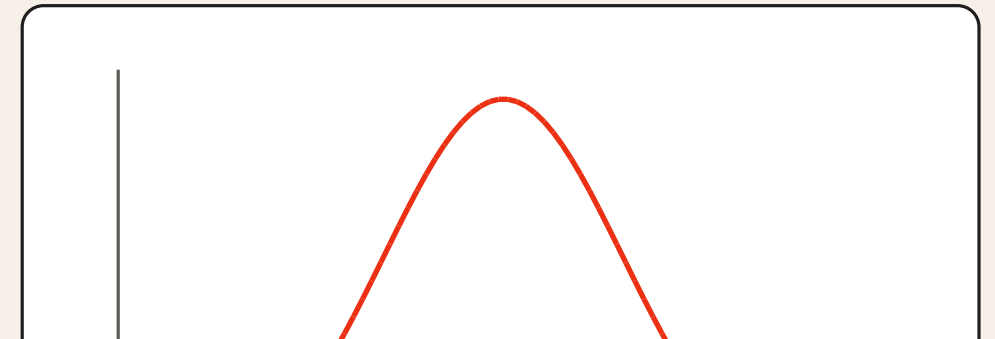
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CLASS TOPIC

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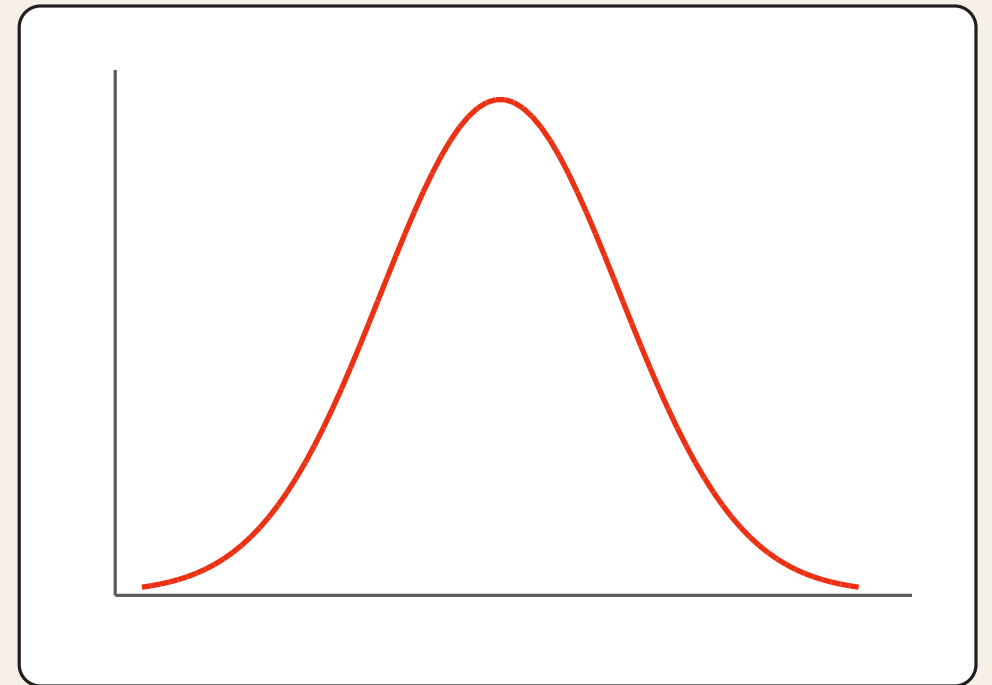
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CLASS TOPIC

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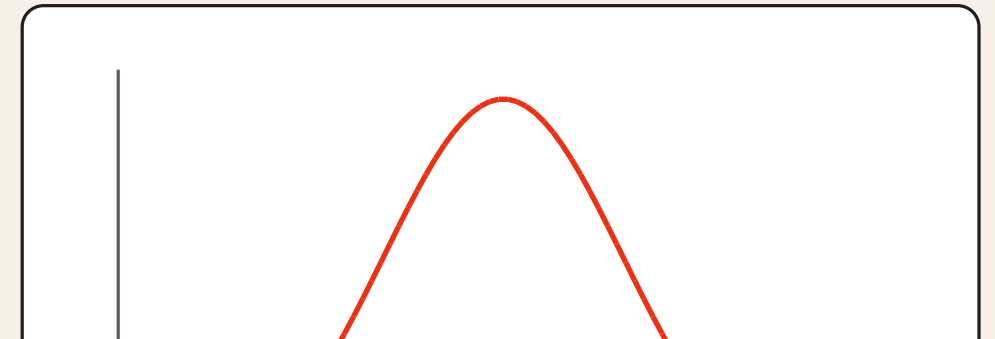
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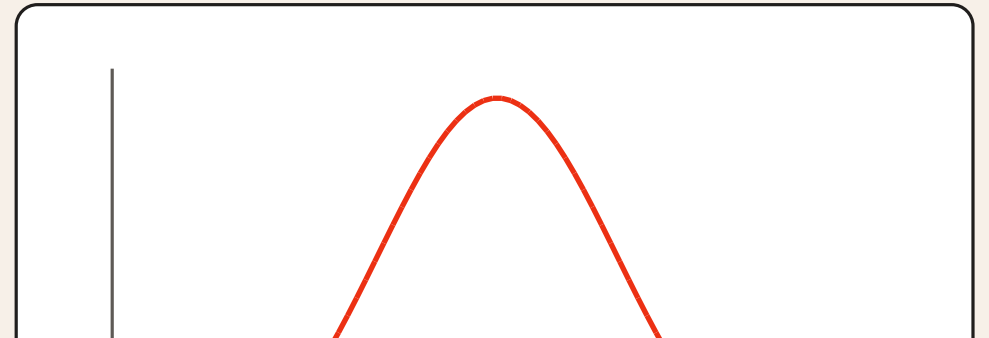
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FORMULA

$$E(\bar{X}) = \mu$$

$$SE(\bar{X}) = \sigma/\sqrt{n}$$

$$\text{if } \sigma \text{ unknown: } SE(\bar{X}) = s/\sqrt{n}$$

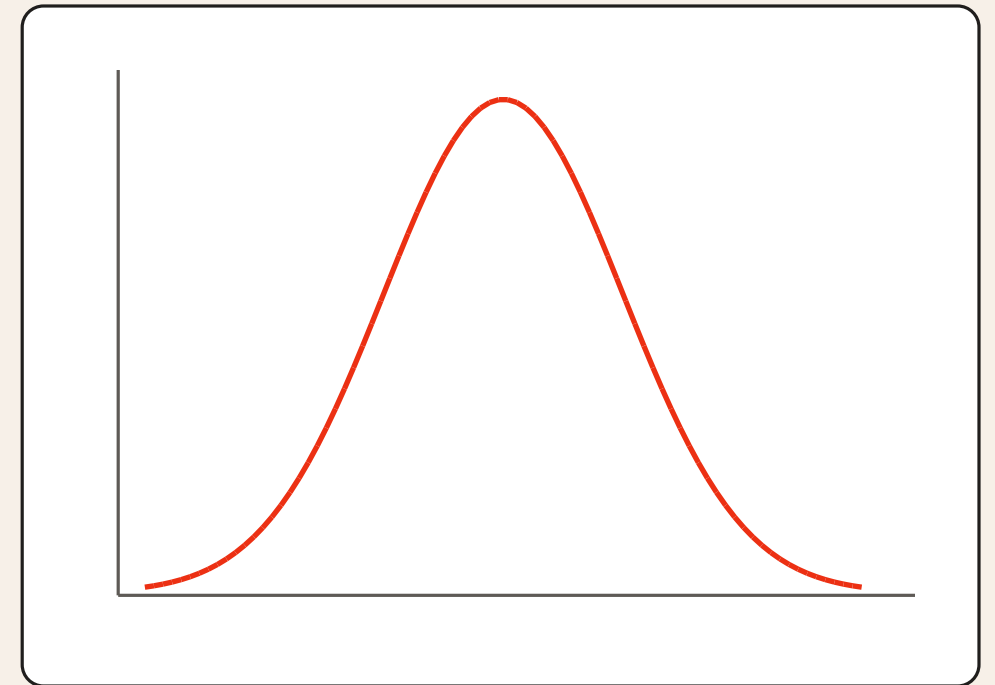


CLASS TOPIC

Sampling distributions and the central limit theorem

How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
- Separate statistical evidence from causal claims.



FORMULA

$$E(X) = \mu$$

CLASS TOPIC

Sampling distributions and the central limit theorem

Final checklist

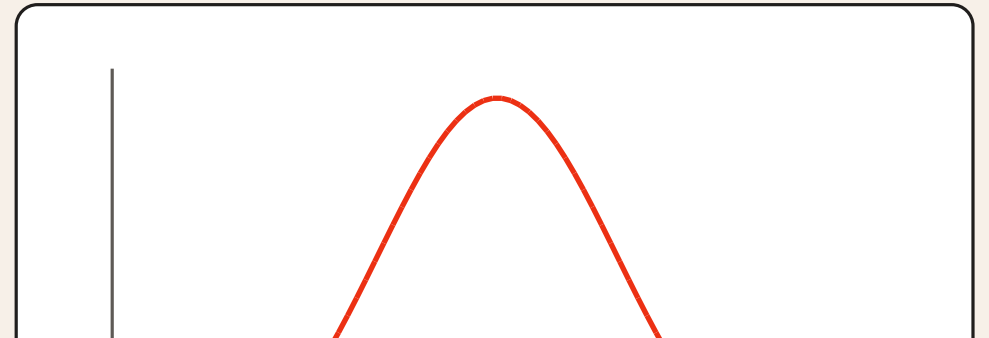
- Question is clear.
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CLASS TOPIC

Point and interval estimation

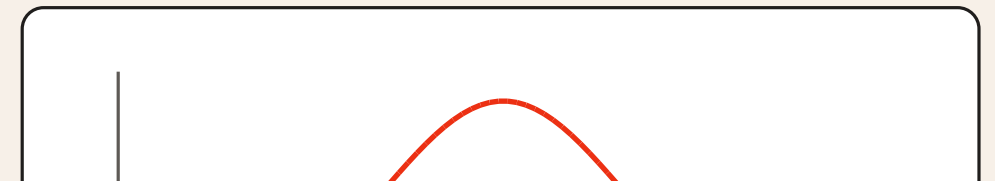
What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
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W09**Statistics**

Clean formulas · aligned layout · corrected te

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



CLASS TOPIC

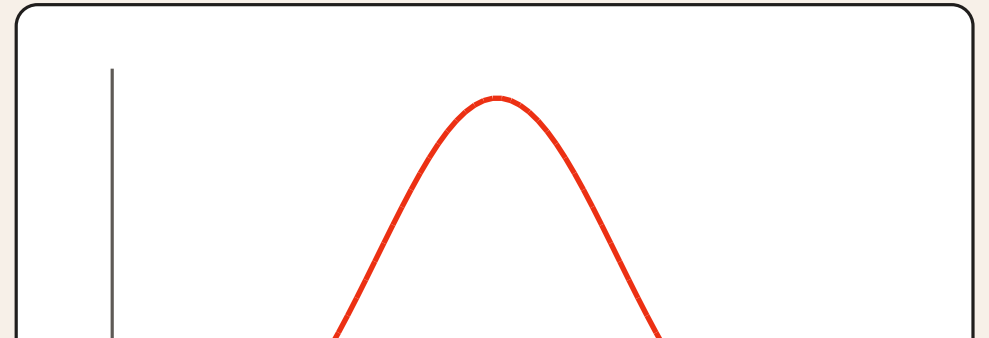
Point and interval estimation

Economic question first

- Start from a real decision: compare, describe, forecast or test.
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FORMULA

estimator $\pm$ margin of error
 $CI = \text{estimate} \pm \text{critical value} \times SE$



CLASS TOPIC

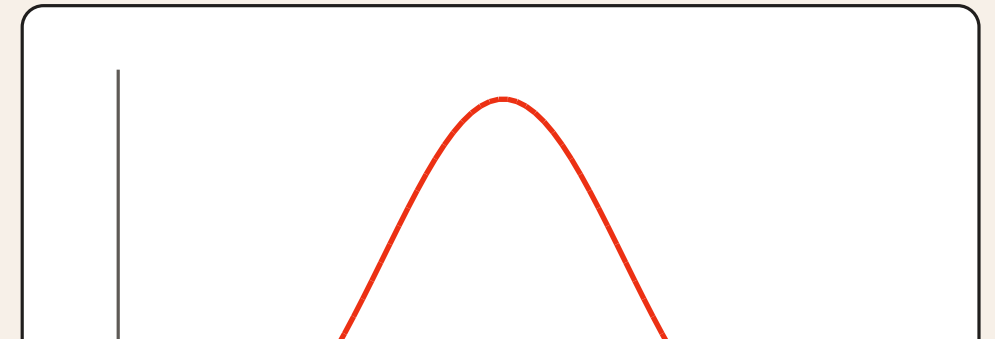
Point and interval estimation

Key definitions

- Population: full group of interest.
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- Statistic: number computed from a sample;
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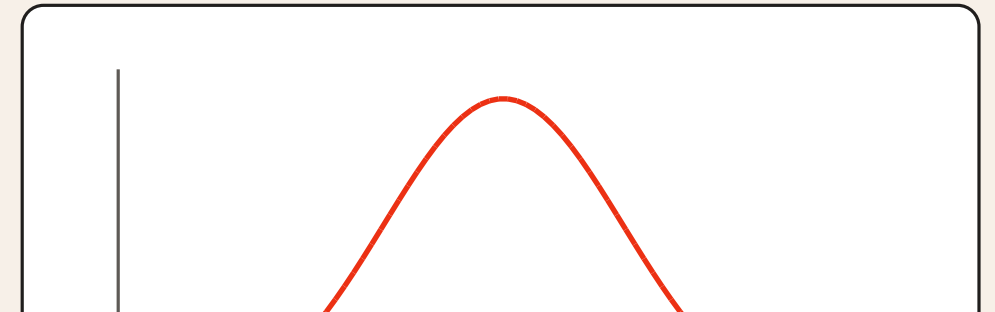
Point and interval estimation

Main formula box

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CLASS TOPIC

Point and interval estimation

Mini example

- Write the data clearly.
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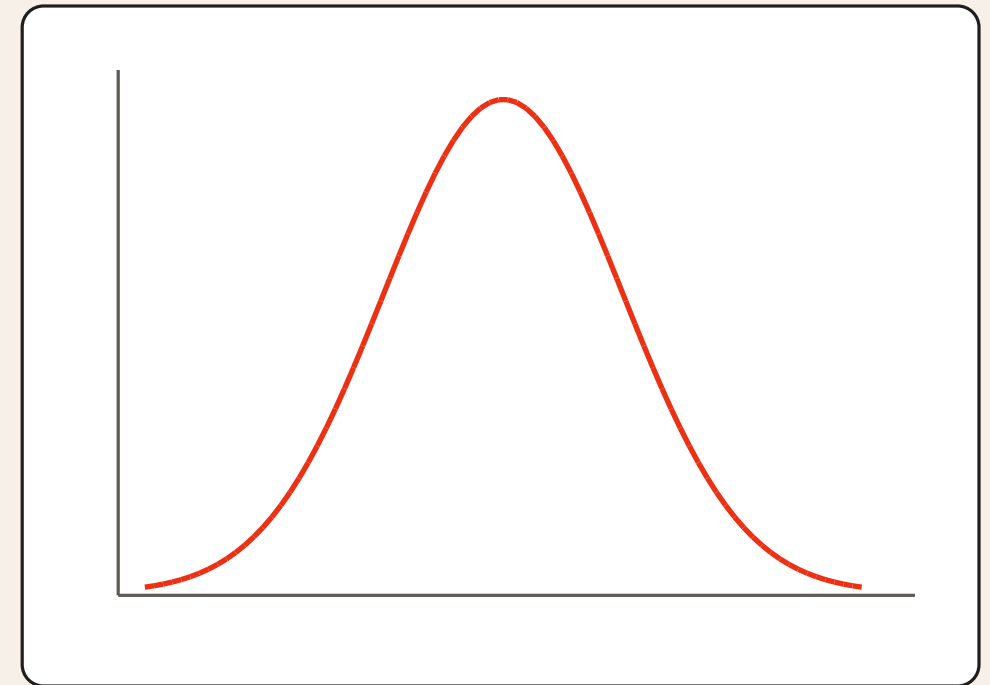
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CLASS TOPIC

Point and interval estimation

Visual intuition

- A graph should answer a question, not decorate the slide.
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- Look for center, spread, shape and unusual values.

**FORMULA**

estimator $\pm$ margin of error

CLASS TOPIC

Point and interval estimation

Second formula box

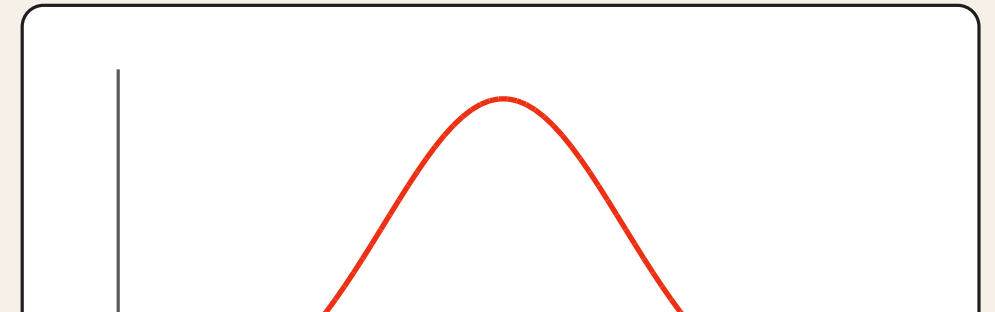
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FORMULA 2

mean: $\bar{x} \pm t^* \cdot s/\sqrt{n}$

proportion: $\hat{p} \pm z^* \sqrt{[\hat{p}(1-\hat{p})/n]}$

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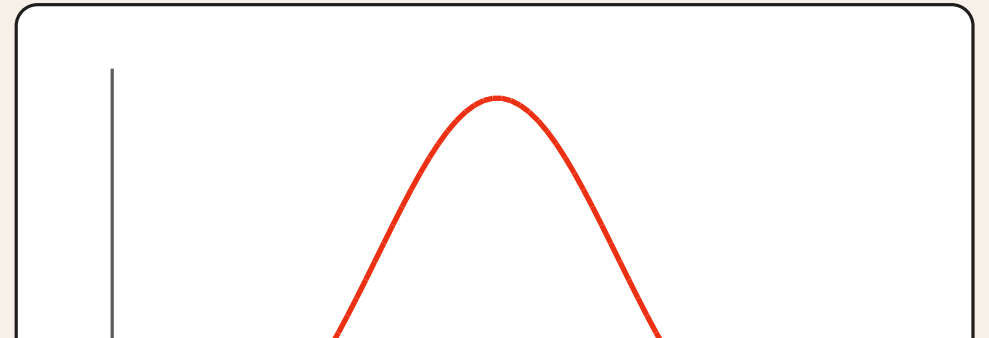
Point and interval estimation

Common mistakes

- Mixing sample and population symbols.
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CLASS TOPIC

Point and interval estimation

R / spreadsheet check

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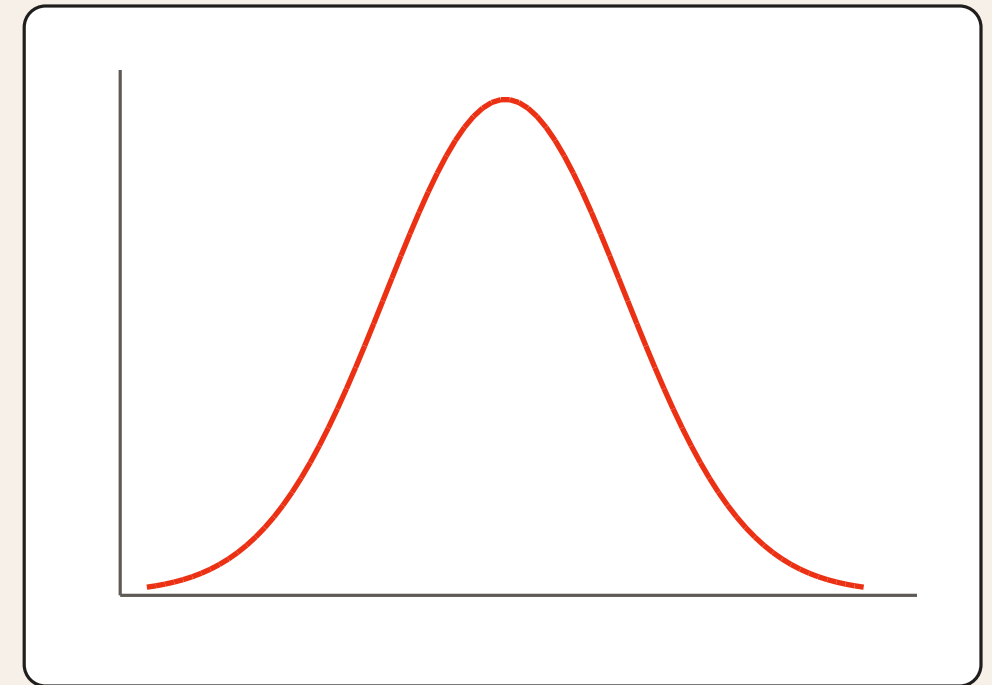
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CLASS TOPIC

Point and interval estimation

Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.

**FORMULA**

estimator $\pm$ margin of error

CLASS TOPIC

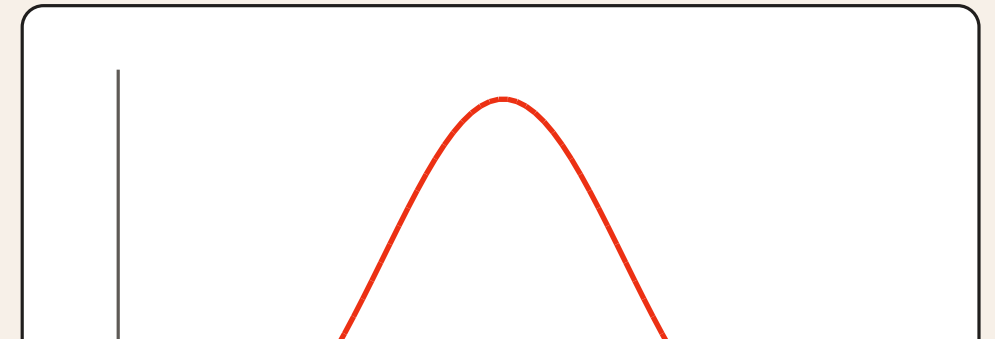
Point and interval estimation

Conditions and assumptions

- Check whether the method fits the data type.
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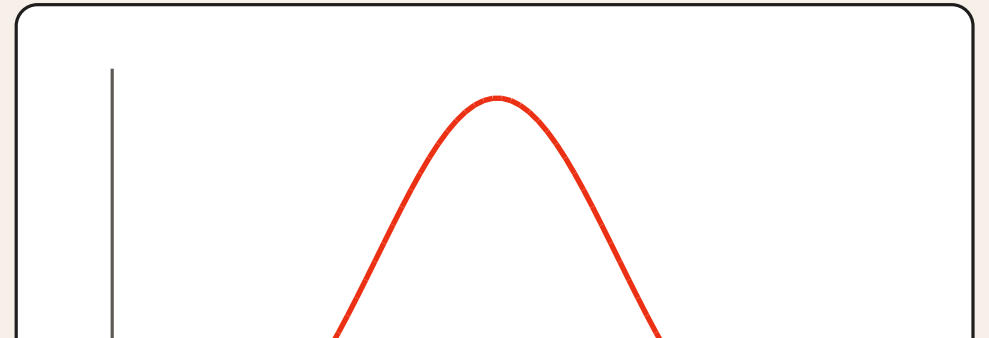
Point and interval estimation

Class exercise

- Solve one calculation by hand.
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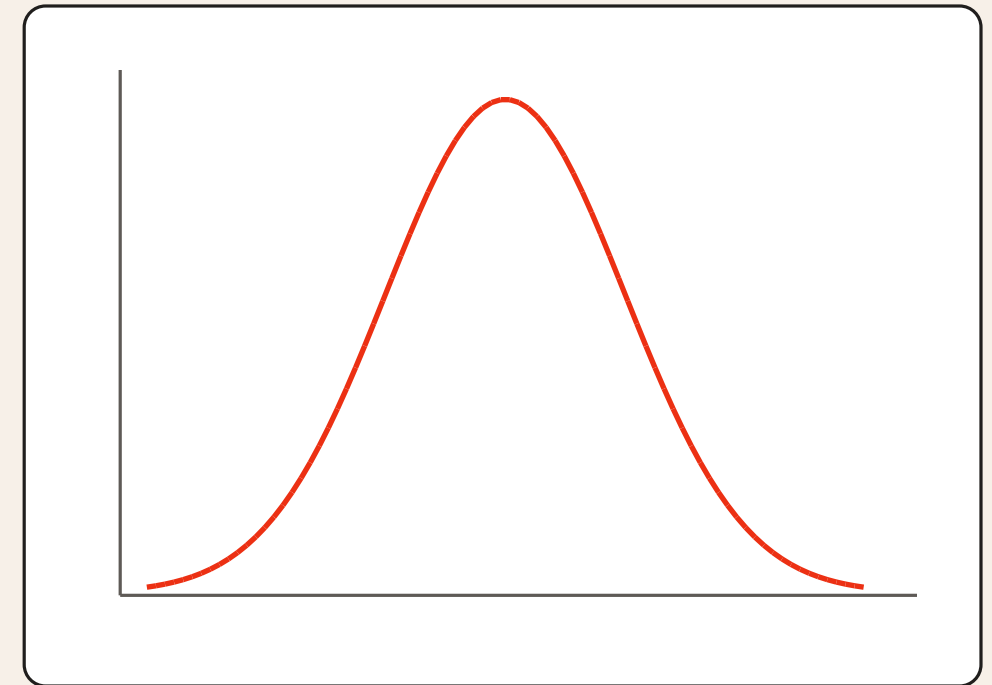


CLASS TOPIC

Point and interval estimation

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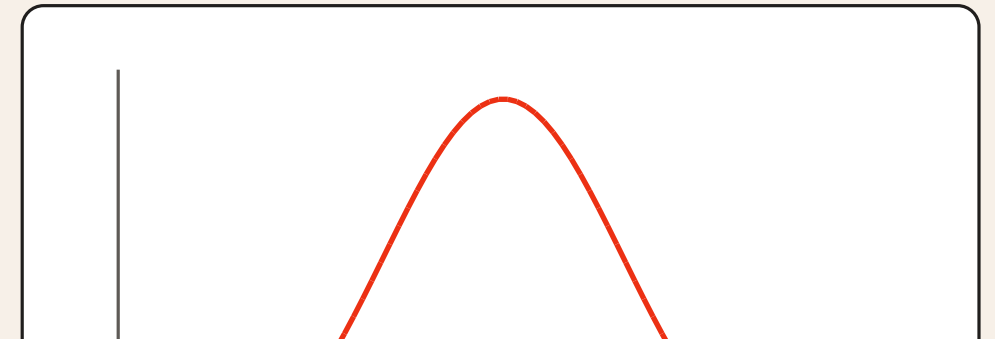
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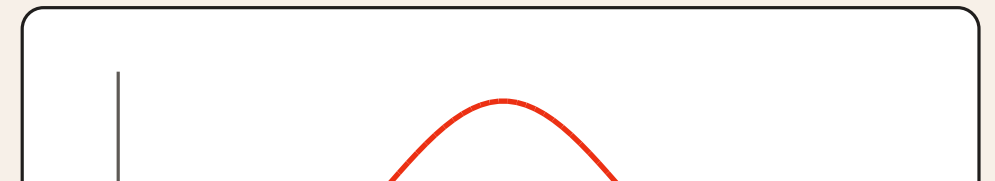
Hypothesis testing I - means and proportions

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W10**Statistics**

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CLASS TOPIC

Hypothesis testing I - means and proportions

Economic question first

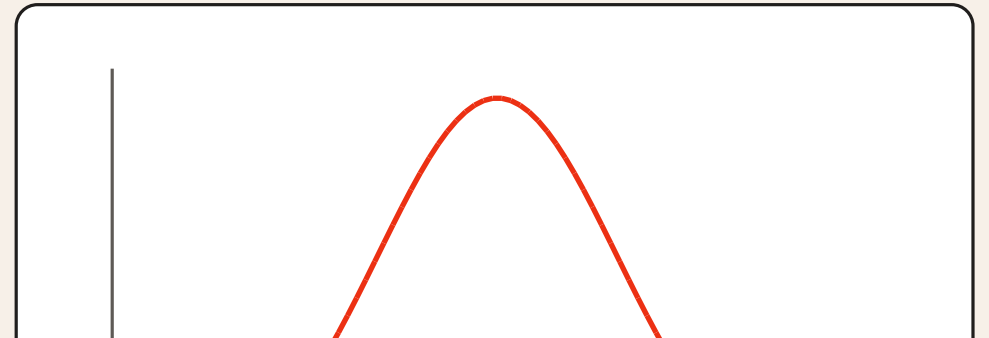
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FORMULA

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0 \text{ (or } >, < \text{)}$$

$$\text{test statistic} = (\text{estimate} - \text{null value}) / \text{SE}$$



CLASS TOPIC

Hypothesis testing I - means and proportions

Key definitions

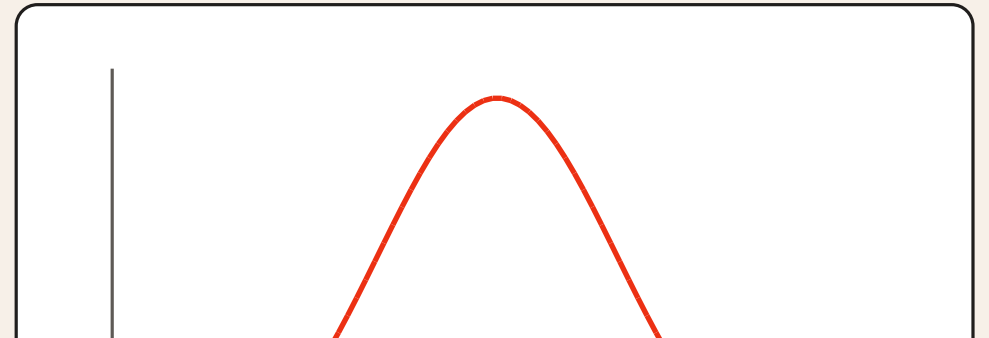
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CLASS TOPIC

Hypothesis testing I - means and proportions

Main formula box

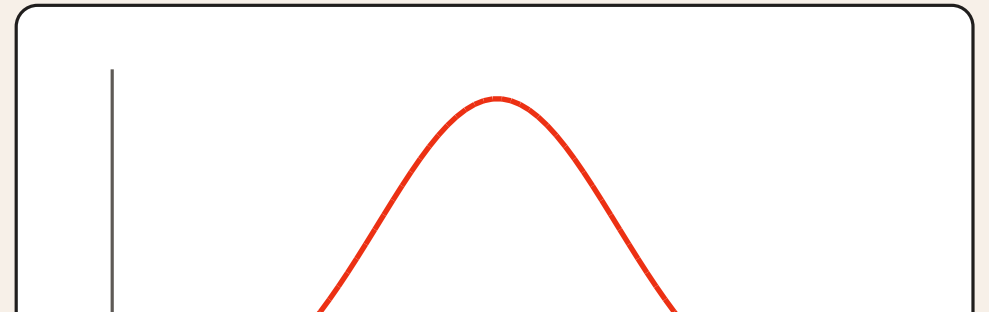
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CLASS TOPIC

Hypothesis testing I - means and proportions

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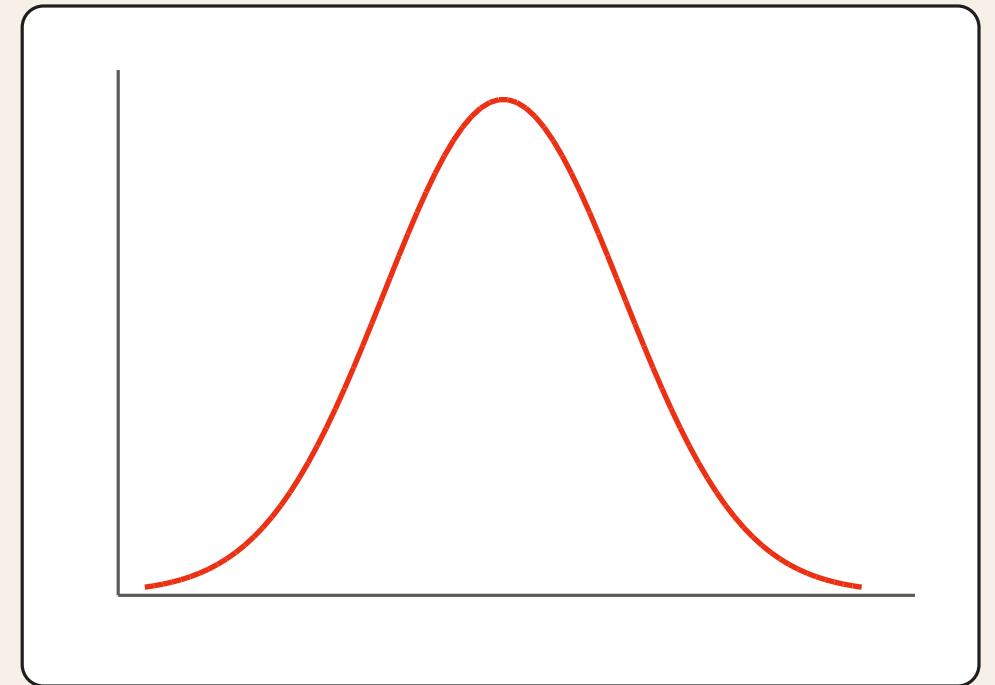
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CLASS TOPIC

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CLASS TOPIC

Hypothesis testing I - means and proportions

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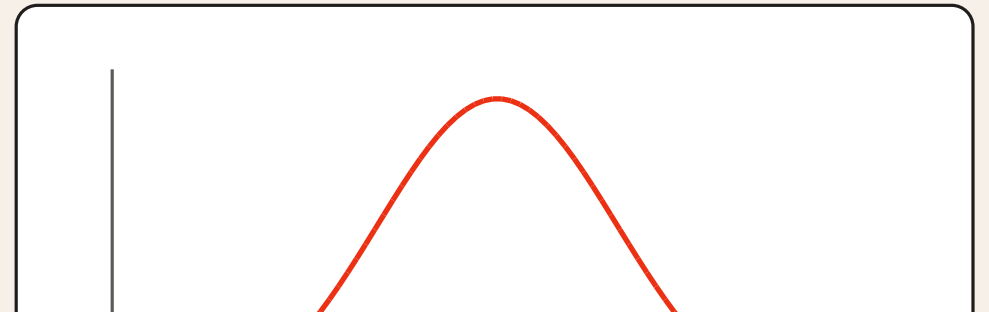
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FORMULA 2

$$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$$

$$z = (\hat{p} - p_0)/\sqrt{[p_0(1-p_0)/n]}$$

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



CLASS TOPIC

Hypothesis testing I - means and proportions

Common mistakes

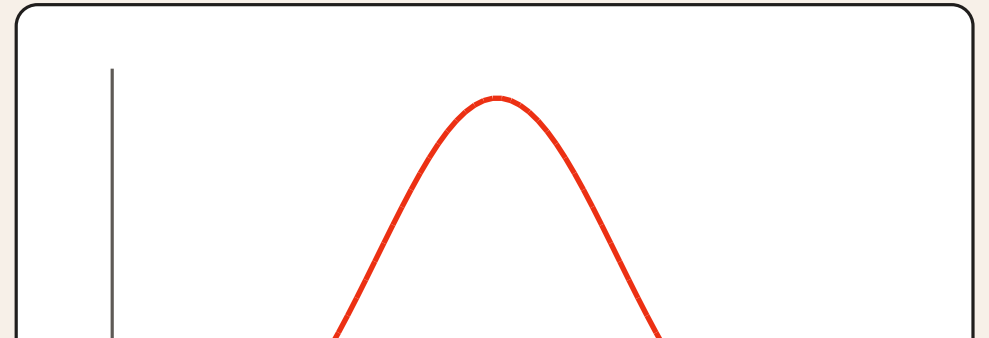
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CLASS TOPIC

Hypothesis testing I - means and proportions

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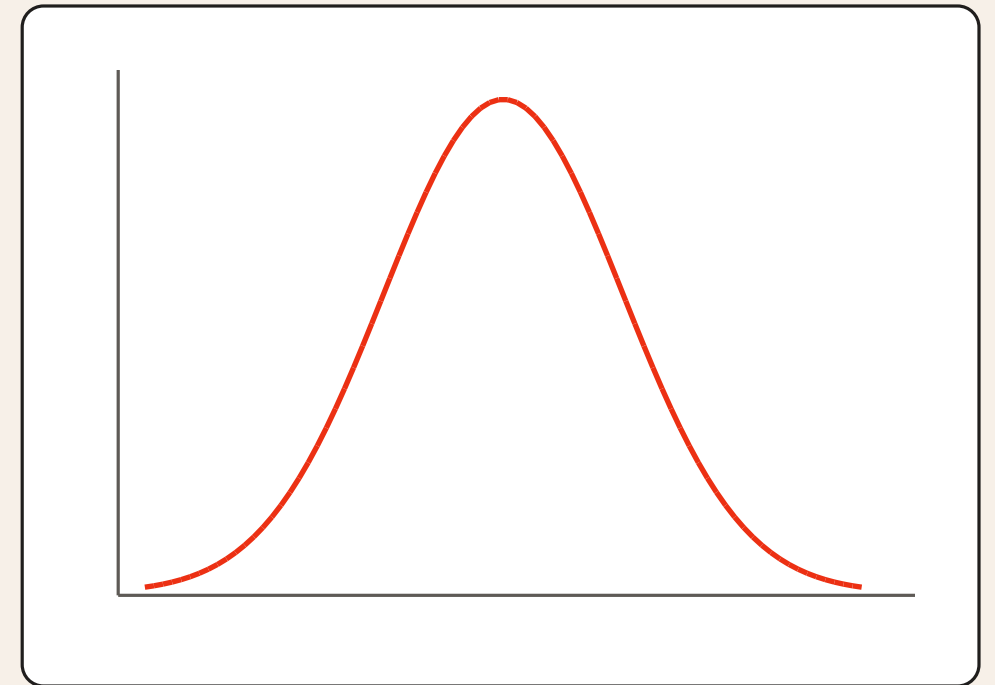
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CLASS TOPIC

Hypothesis testing I - means and proportions

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CLASS TOPIC

Hypothesis testing I - means and proportions

Conditions and assumptions

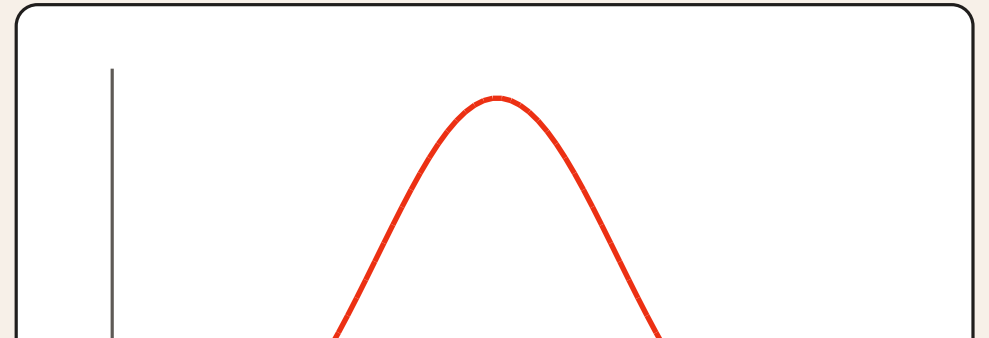
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CLASS TOPIC

Hypothesis testing I - means and proportions

Class exercise

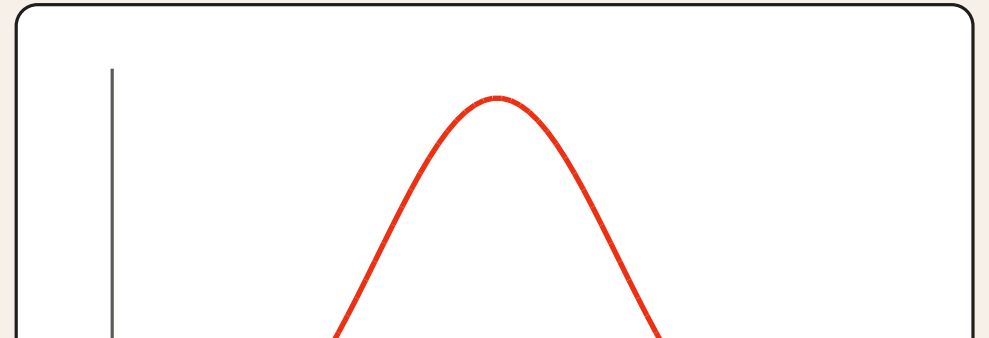
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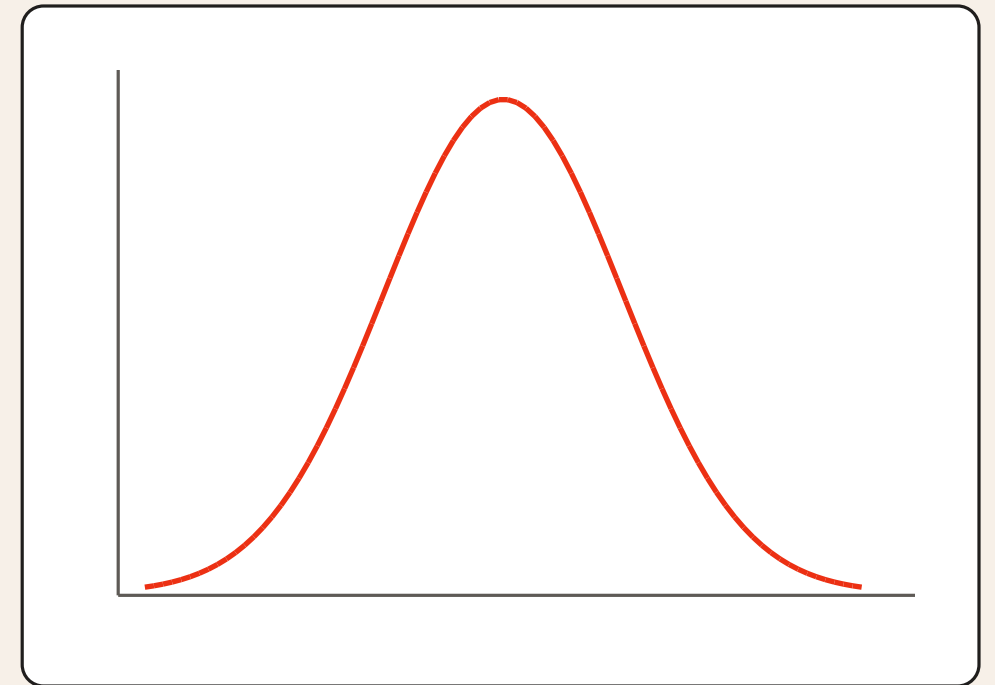


CLASS TOPIC

Hypothesis testing I - means and proportions

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CLASS TOPIC

Hypothesis testing I - means and proportions

Final checklist

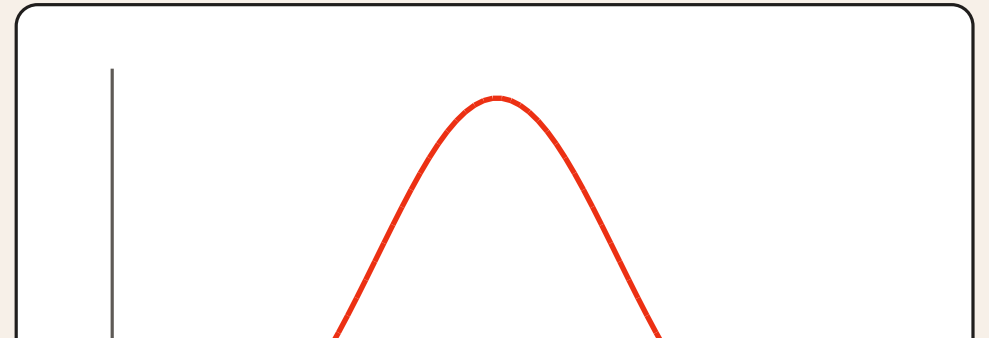
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CLASS TOPIC

Hypothesis testing II - two samples and chi-square

What this class fixes

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W11**Statistics**

Clean formulas · aligned layout · corrected te

CLASS TOPIC

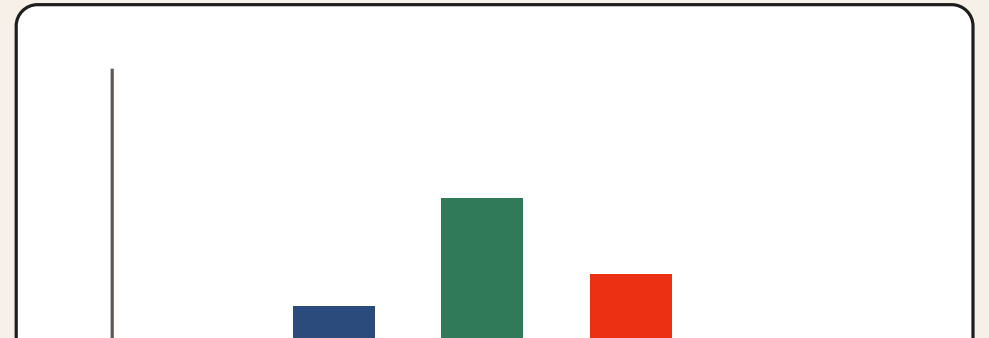
Hypothesis testing II - two samples and chi-square

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CLASS TOPIC

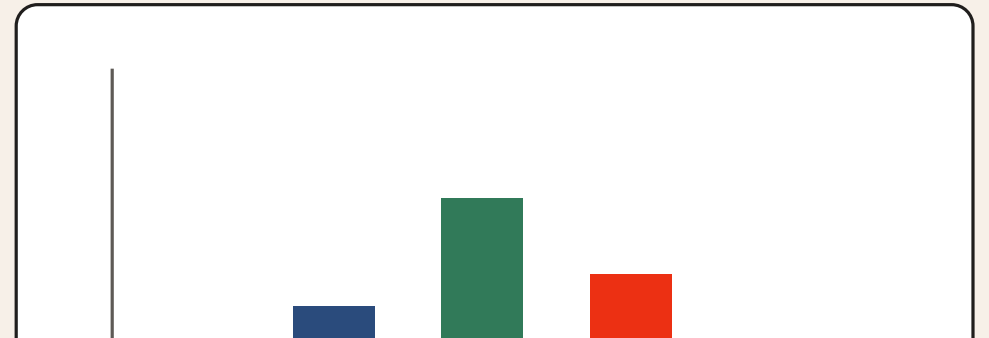
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CLASS TOPIC

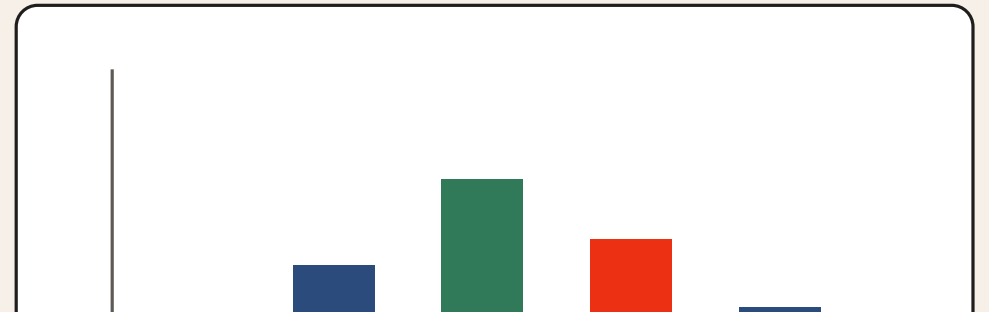
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CLASS TOPIC

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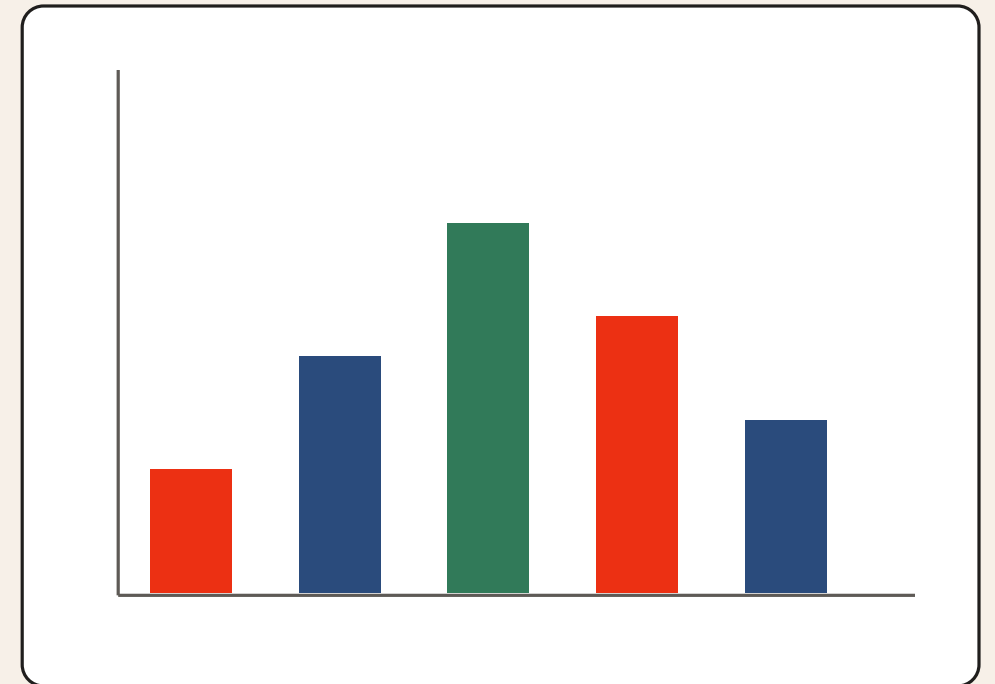
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CLASS TOPIC

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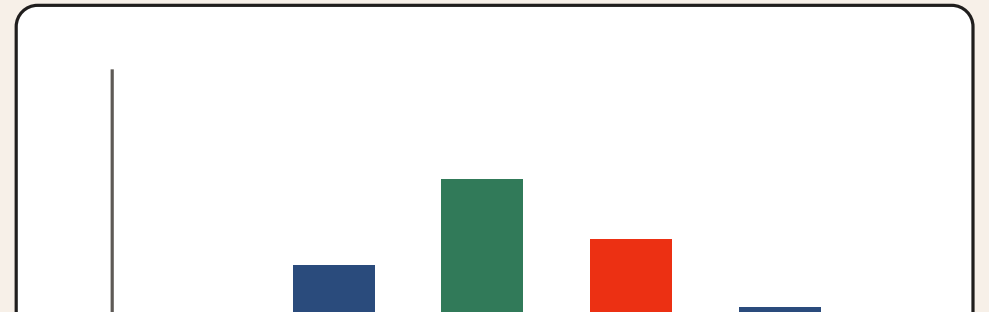
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FORMULA 2

$$\chi^2 = \sum (\text{Observed} - \text{Expected})^2 / \text{Expected}$$
$$\text{df} = (\text{rows} - 1)(\text{columns} - 1)$$

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CLASS TOPIC

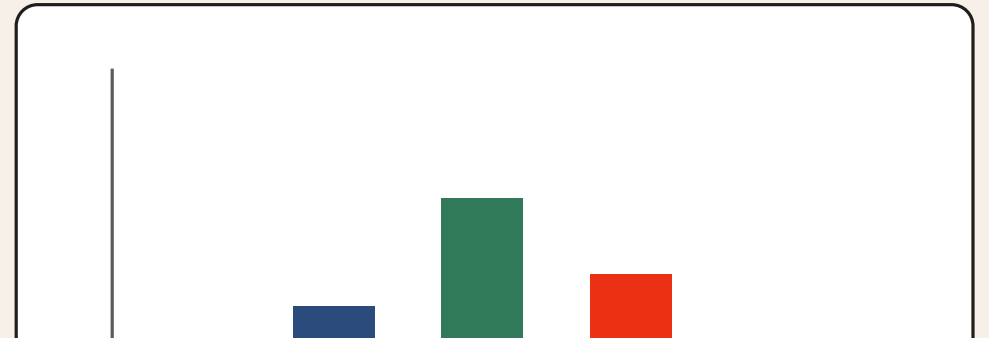
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CLASS TOPIC

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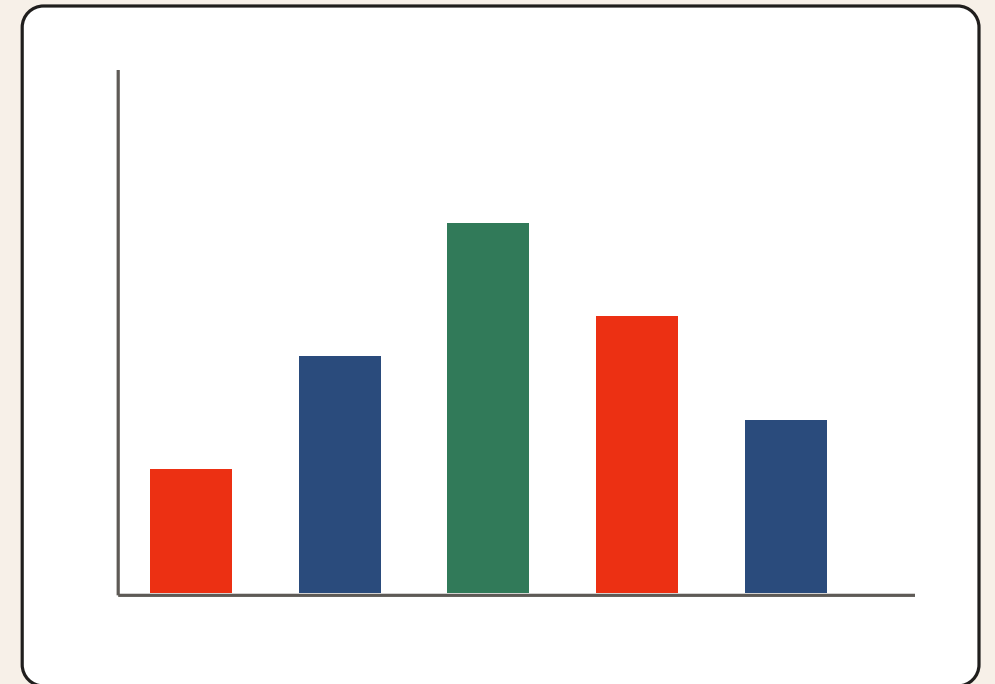
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CLASS TOPIC

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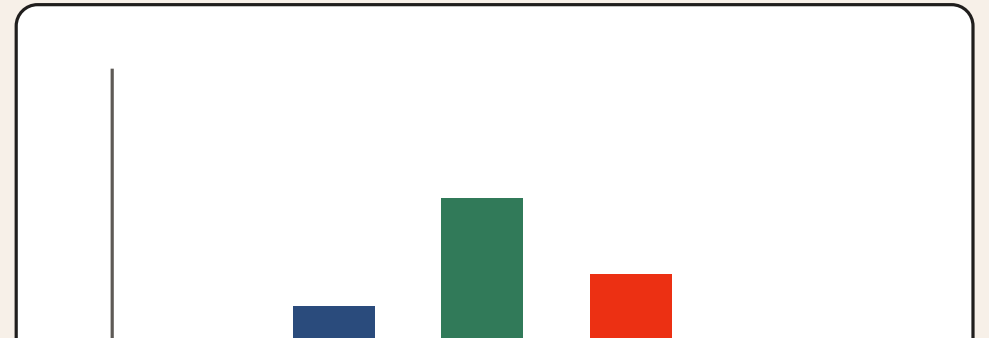
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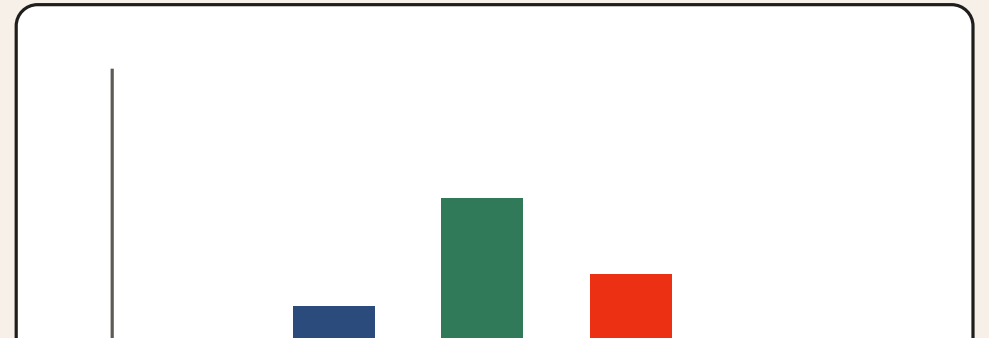
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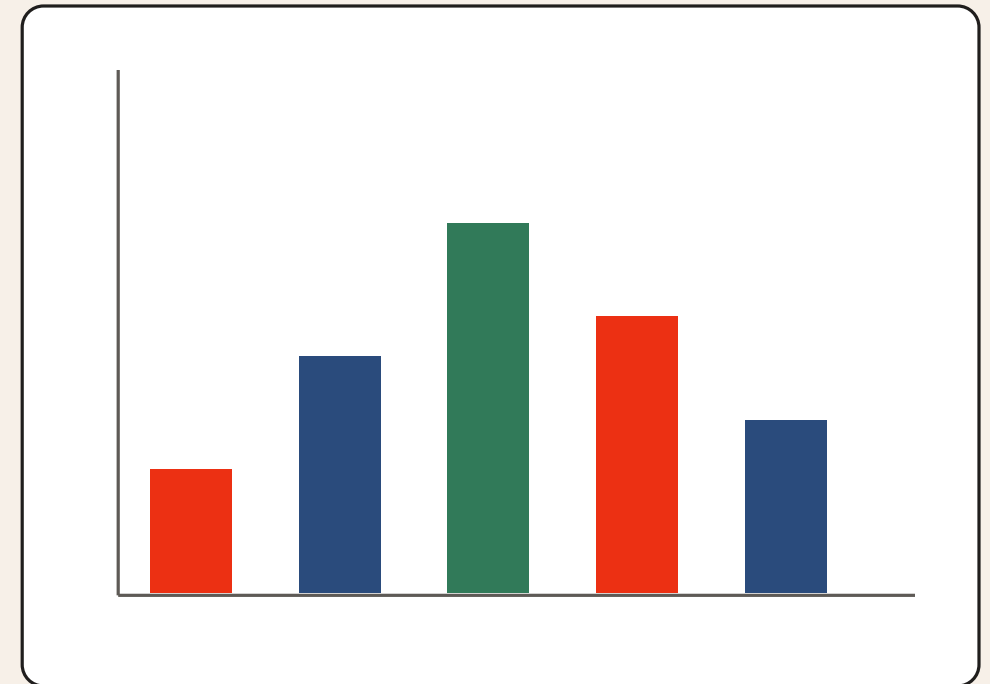


CLASS TOPIC

Hypothesis testing II - two samples and chi-square

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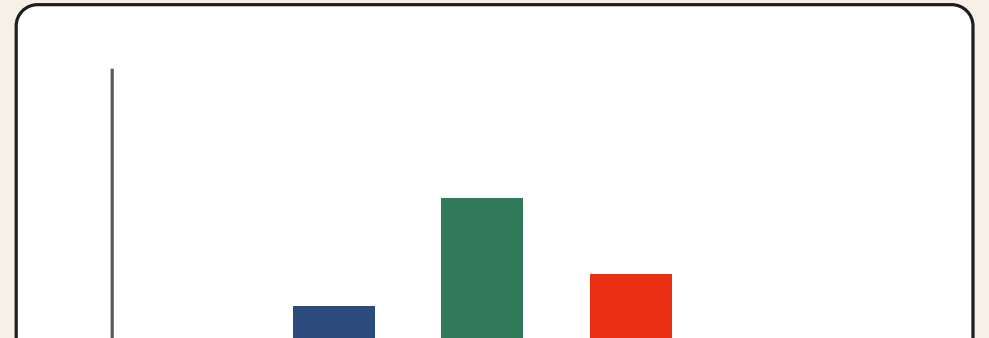
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CLASS TOPIC

Simple linear regression and residual diagnostics

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W12**Statistics**

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CLASS TOPIC

Simple linear regression and residual diagnostics

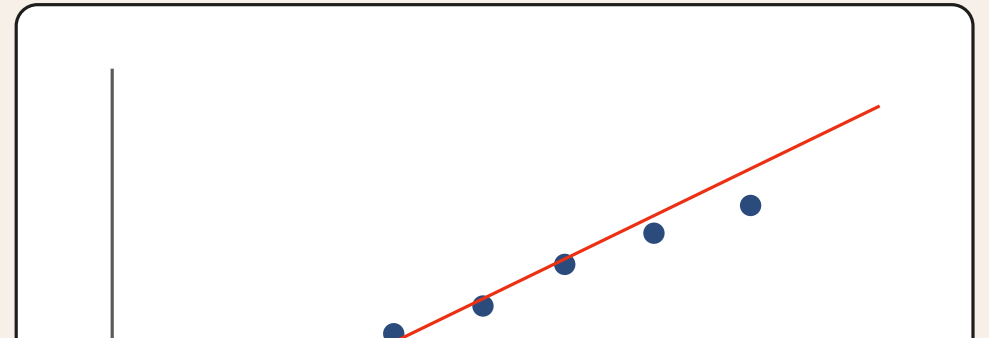
Economic question first

- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$



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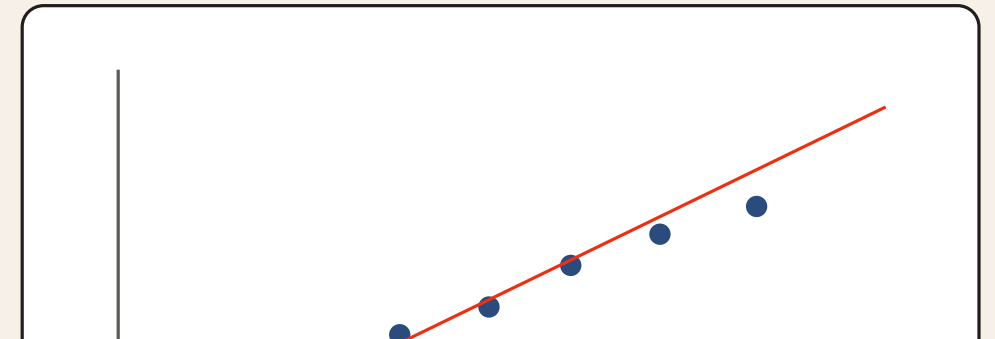
Key definitions

- Population: full group of interest.
- Sample: observed part of the population.
- Statistic: number computed from a sample;
parameter: number for the population.

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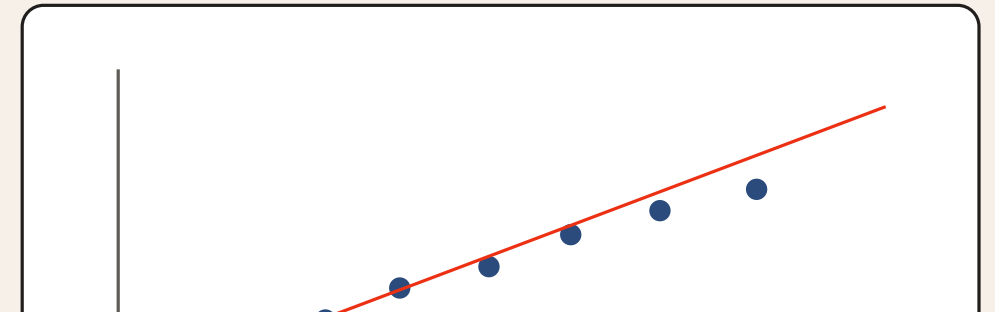
Main formula box

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- Read every symbol before calculating.
- Use parentheses carefully.

FORMULA 1

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CLASS TOPIC

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Mini example

- Write the data clearly.
- Substitute values step by step.
- Round only at the final reporting stage.

01 Data: 4, 7, 8, 9, 12

02 Step 1: compute the needed statistic

03 Step 2: write the result with units.

04 Step 3: explain it in one sentence.

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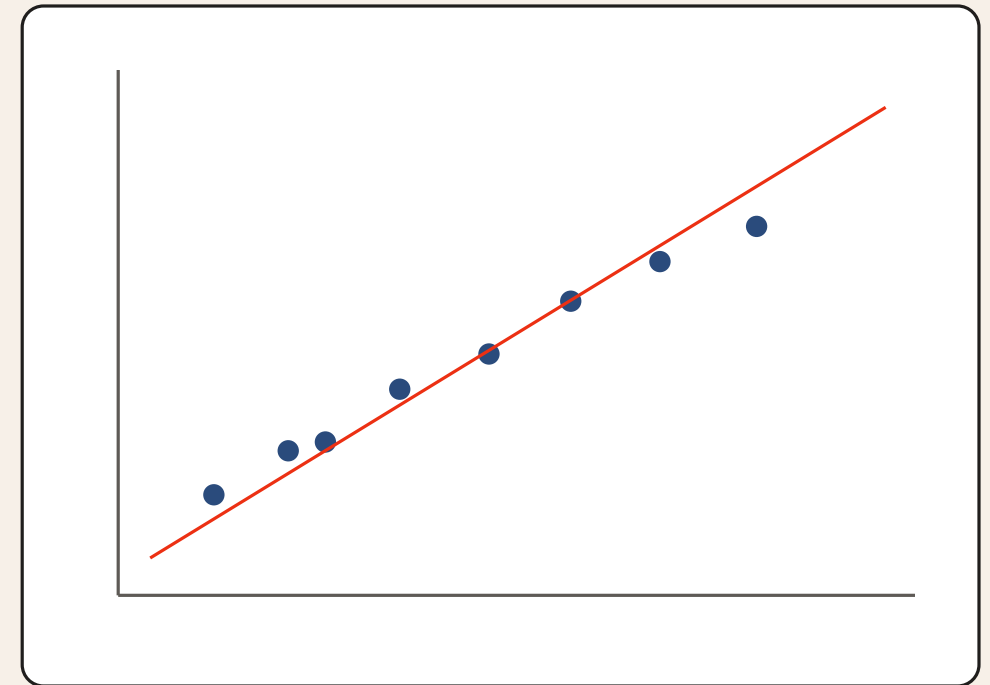
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CLASS TOPIC

Simple linear regression and residual diagnostics

Visual intuition

- A graph should answer a question, not decorate the slide.
- Always label axes and units.
- Look for center, spread, shape and unusual values.



FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

CLASS TOPIC

Simple linear regression and residual diagnostics

Second formula box

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- No broken equation fragments or floating text boxes.
- Notation is consistent across English and Russian versions.

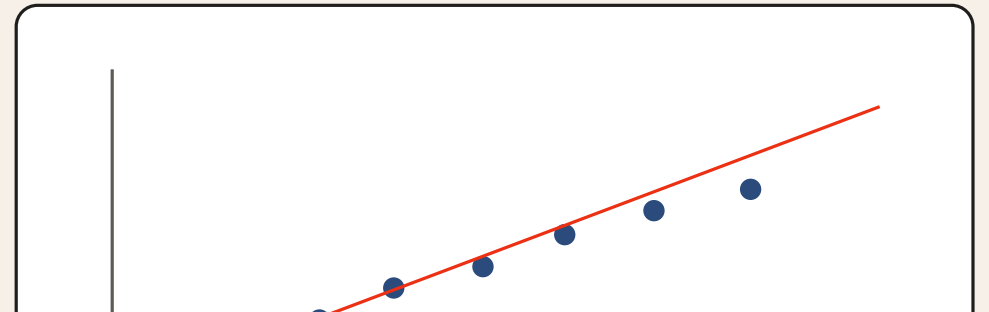
FORMULA 2

$$e_i = y_i - \hat{y}_i$$

$$SSE = \sum e_i^2$$

OLS chooses b_0, b_1 to minimize SSE

Formula boxes fixed: one merged box, consistent symbols, no broken fragments.



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Simple linear regression and residual diagnostics

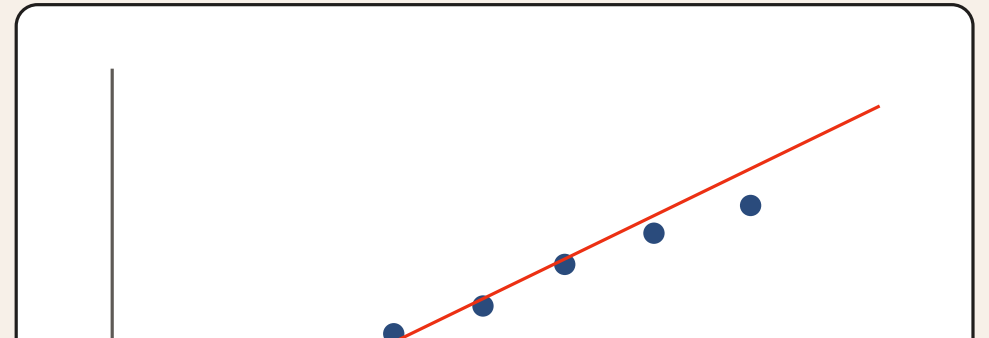
Common mistakes

- Mixing sample and population symbols.
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CLASS TOPIC

Simple linear regression and residual diagnostics

R / spreadsheet check

- Use software to calculate, but keep the statistical logic visible.
- Check variable names, missing values and units first.
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R / EXCEL

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# example pattern  
mean(x); sd(x); summary(x)  
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$$y_i = \beta_0 + \beta_1 x_i + u_i$$

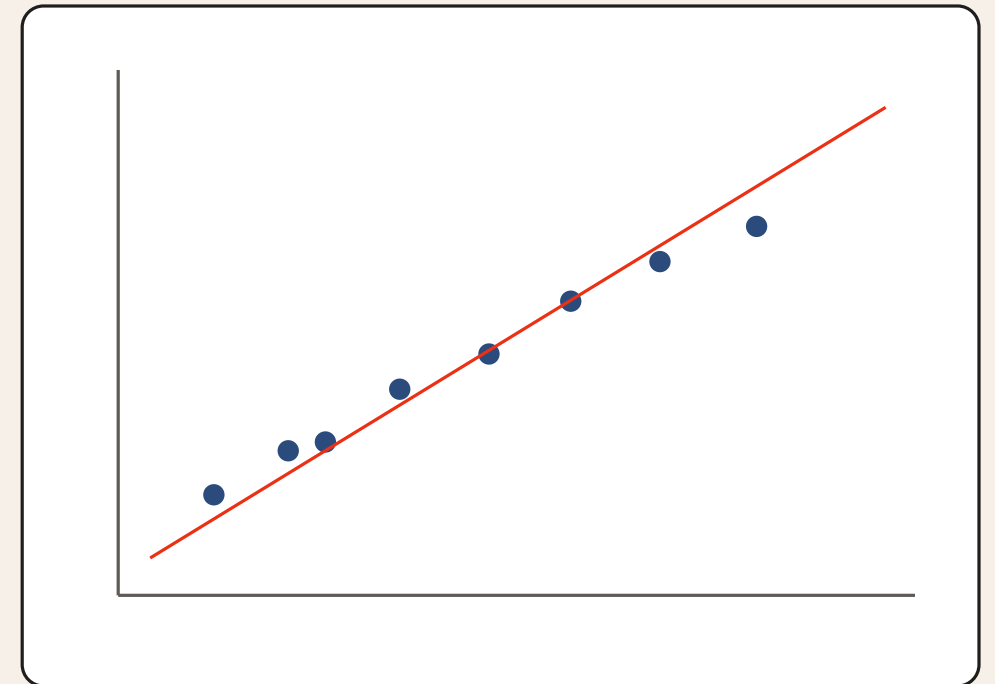
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CLASS TOPIC

Simple linear regression and residual diagnostics

Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.



FORMULA

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

CLASS TOPIC

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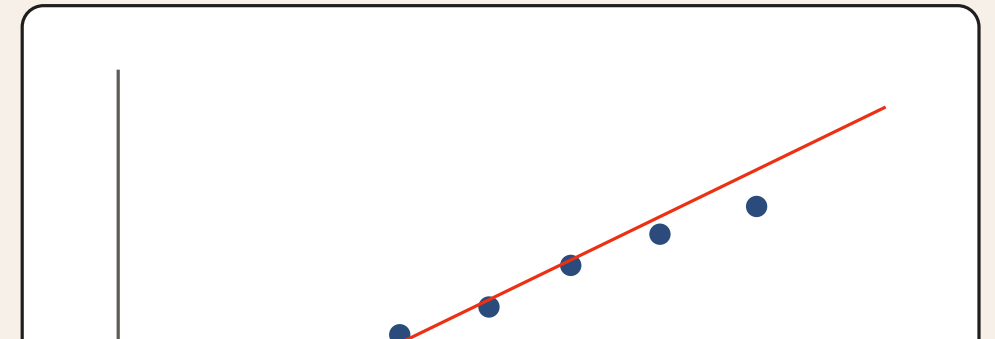
Conditions and assumptions

- Check whether the method fits the data type.
- Check independence and sample size when needed.
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CLASS TOPIC

Simple linear regression and residual diagnostics

Class exercise

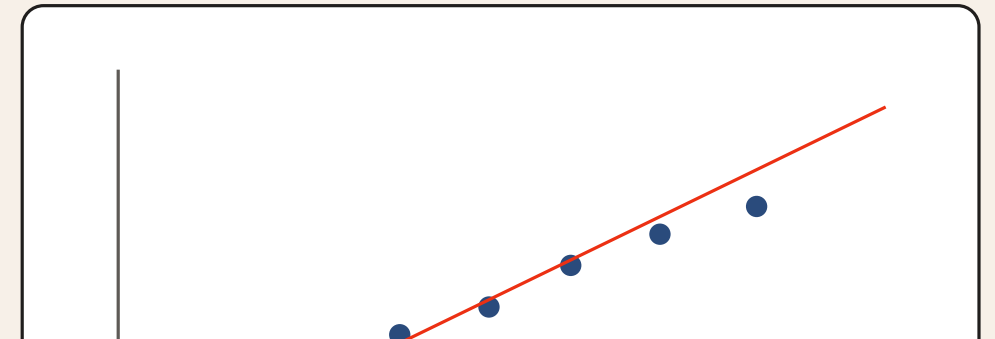
- Solve one calculation by hand.
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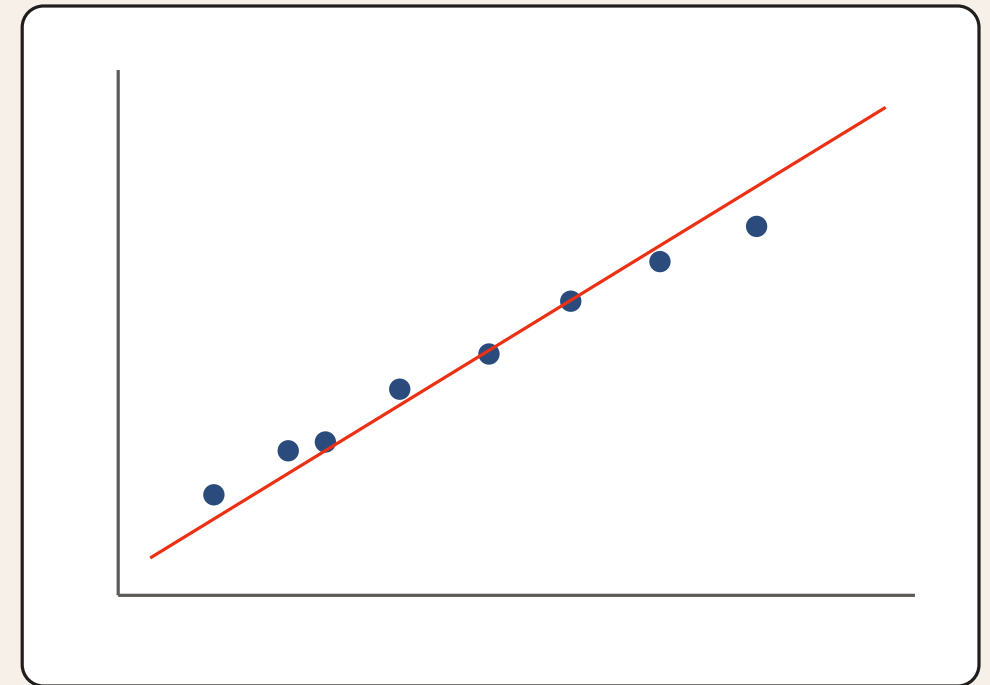


CLASS TOPIC

Simple linear regression and residual diagnostics

How to report

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CLASS TOPIC

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Final checklist

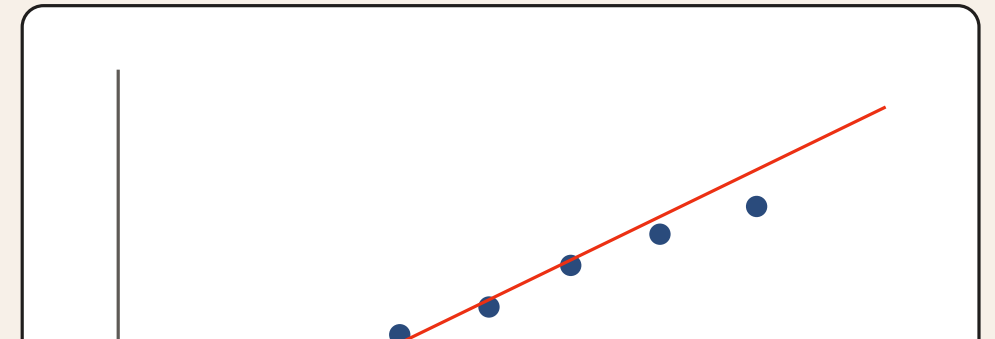
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CLASS TOPIC

Inference in regression; correlation vs causation

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

W13**Statistics**

Clean formulas · aligned layout · corrected te



CLASS TOPIC

Inference in regression; correlation vs causation

Economic question first

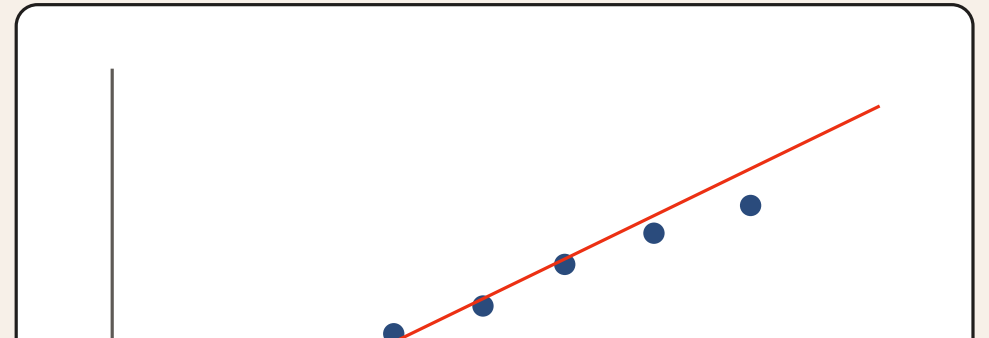
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CLASS TOPIC

Inference in regression; correlation vs causation

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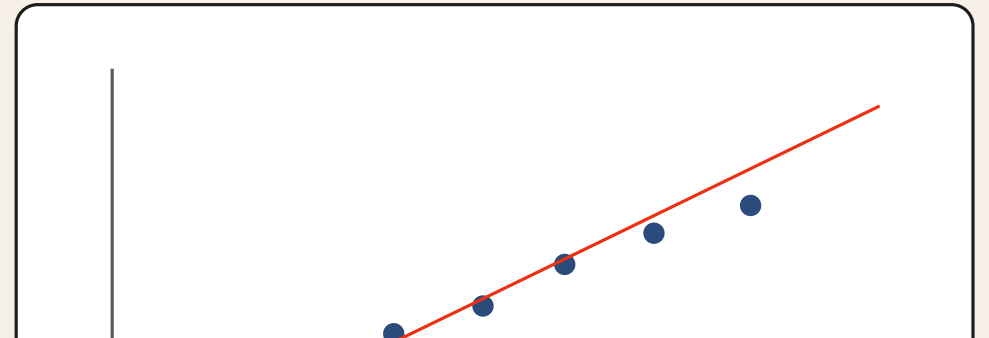
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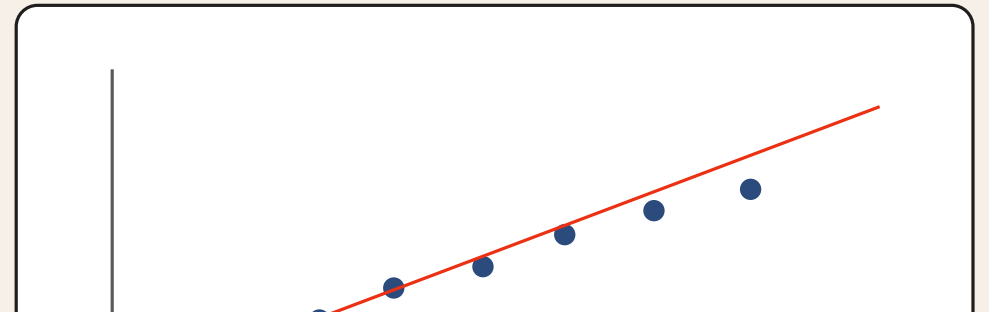
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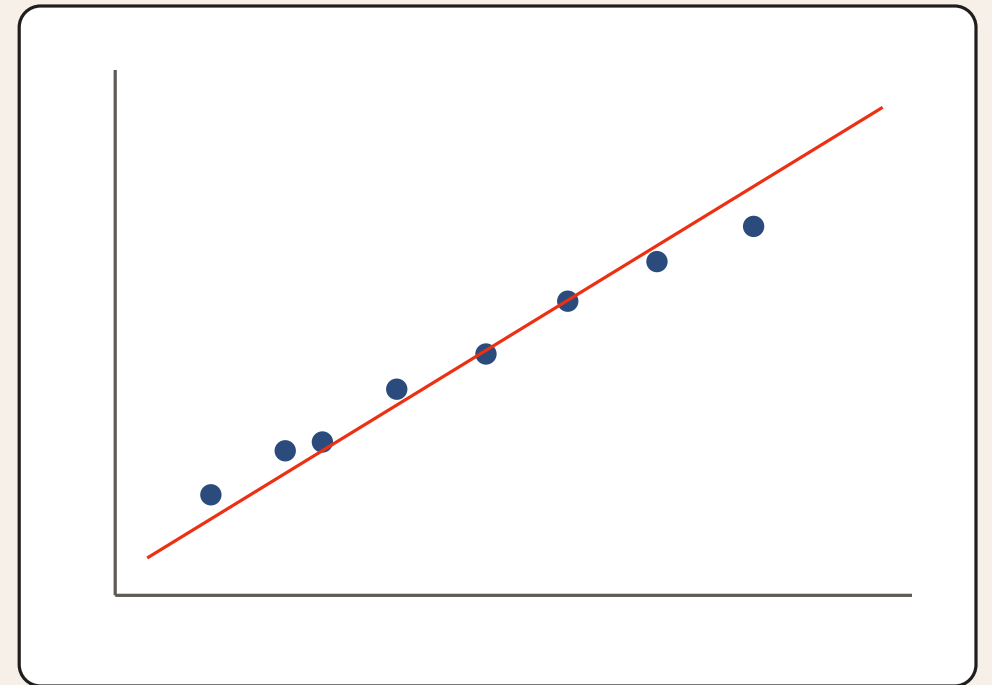
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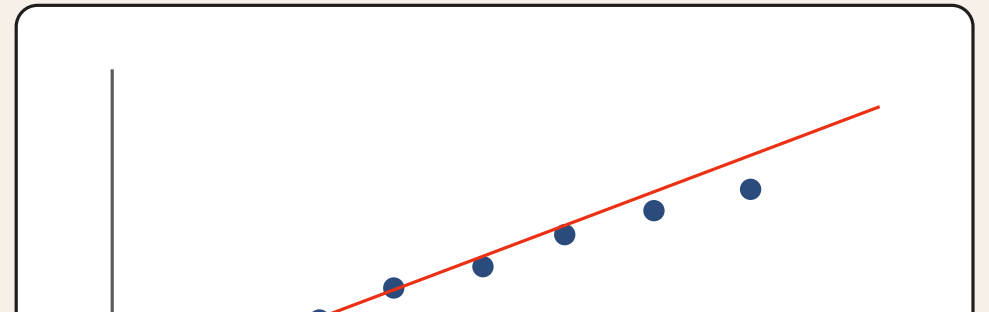
FORMULA 2

$\text{Corr}(X,Y) \neq \text{causal effect}$

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + u$$

omit $Z \rightarrow$ possible bias in β_1

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CLASS TOPIC

Inference in regression; correlation vs causation

Common mistakes

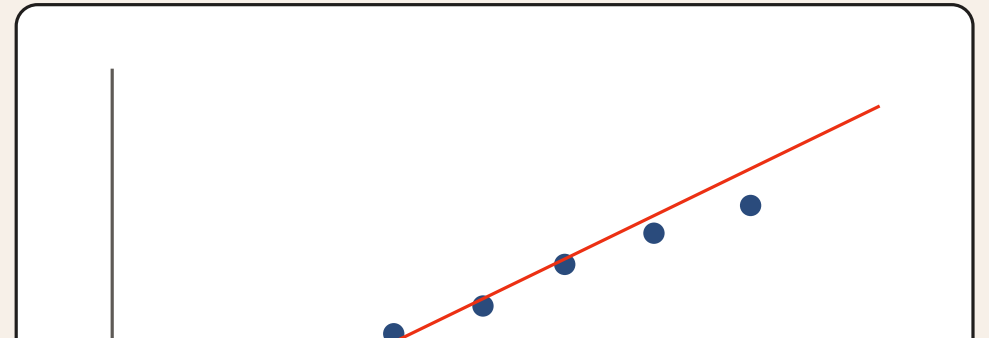
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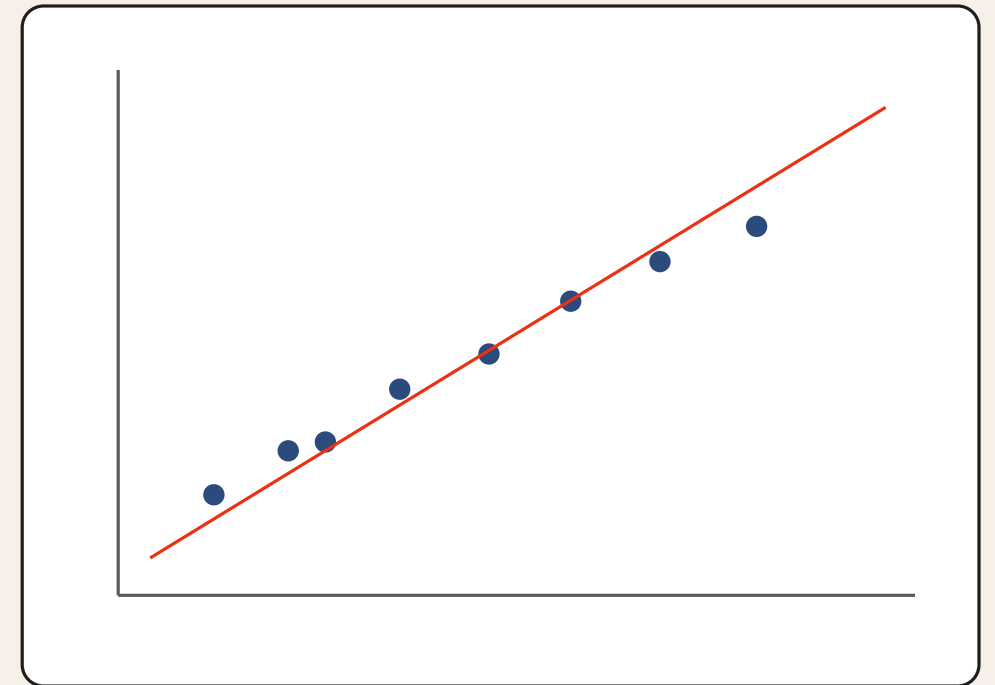
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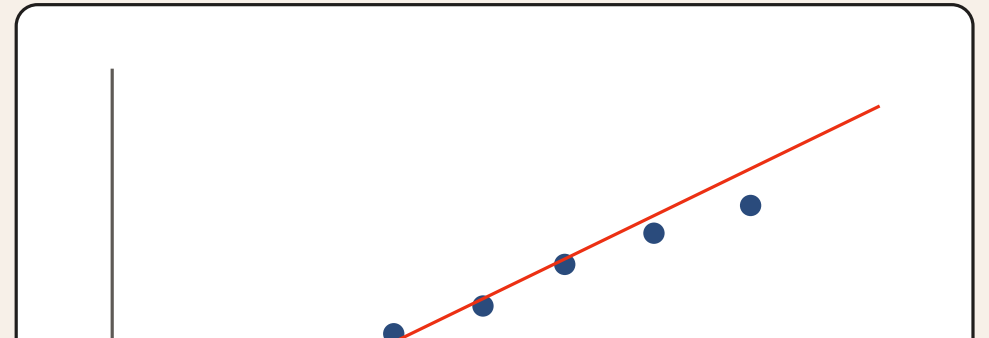
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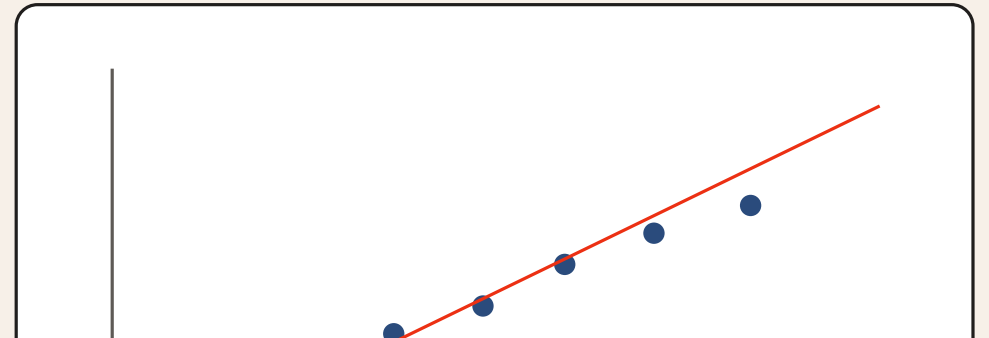
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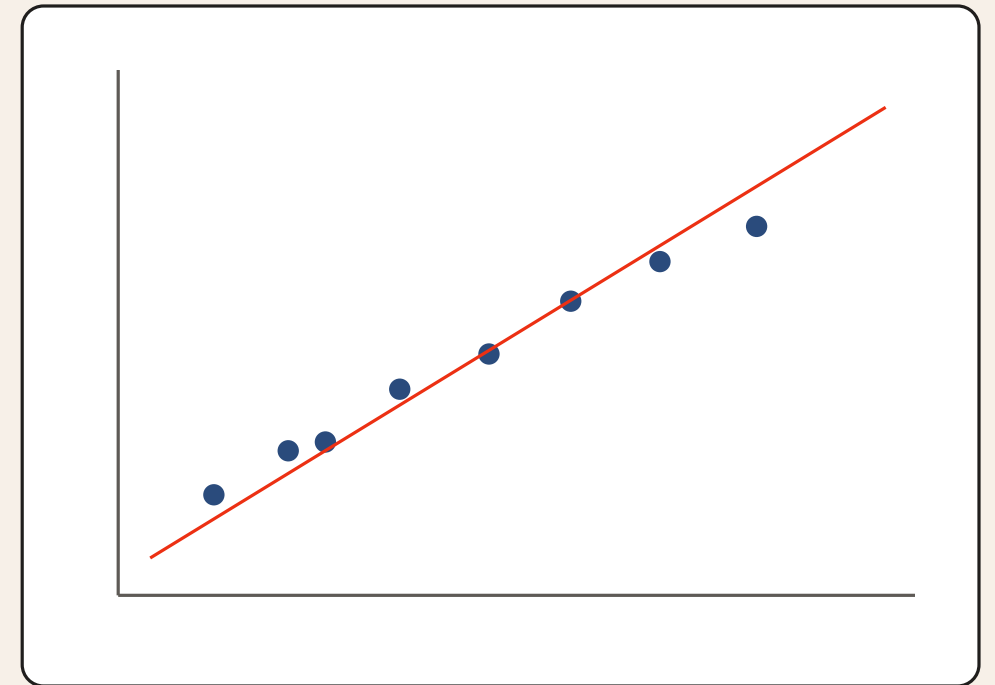


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